Stochastic Ordering of Random Menus

Carlo Andreatta* Efe A. Ok†

May 23, 2026

Abstract

We introduce a notion of first-order stochastic dominance for random feasible sets (= menus) in a suitably ordered alternative space, by using indirect (Bernoulli) utilities. This notion generalizes the classical stochastic ordering of random variables, and enjoys similar characterizations and properties, albeit, now in terms of the Choquet integral instead of the Lebesgue integral. We provide several illustrations that demonstrate how our ordering may be used in economic contexts. In particular, we examine how one may use this ordering to make monotone comparative statics in the context of random budget sets, cores of random cooperative games, choice correspondences on random menus, and ex ante profit maximization. We derive various characterizations and properties of our stochastic ordering.

JEL Classification: D81, C44.
Keywords: Random Sets, First-Order Stochastic Dominance, Monotone Comparative Statics.

1 Introduction

Random sets arise naturally in many areas of economics and probability theory. In economic applications, they often emerge as random menus from which choices will eventually be made once the uncertainty resolves. Examples include random budget sets in consumer theory, random cores of coalitional games, random production sets, and random Pareto sets of social choice theory, among numerous others.[1]

*Department of Economics, New York University. Email: ca3141@nyu.edu.

†Department of Economics and the Courant Institute of Mathematical Sciences, New York University. E-mail: efe.ok@nyu.edu.

[1]We will mainly think of random sets as random menus in this paper. However, this is not the only way random sets are encountered in economics. In particular, starting with the pioneering work of Beresteanu, Molchanov, and Molinari (2011,2012), random sets are widely used as primary tools in partially identified econometric models. Moreover, once one models random menus (may these be budget sets or action spaces), various notions of equilibria manifest as random sets.

In this paper, we are interested in developing methods of ordering random menus, and in turn, using these methods for making monotone comparative statics in a variety of contexts. As a primitive, we work with an alternative space X that is suitably metrized and preordered. (For instance, in the context of consumer choice, we may take X as $\mathbb{R}^n$, and as usual, order it componentwise. Many other examples are given in the body of the paper.) First-order stochastic dominance compares X-valued random variables by looking at their expected utilities for every increasing (and continuous) real-valued function on X. When the uncertainty concerns feasible sets, however, one is naturally interested in the maximum attainable utility levels. We thus extend the stochastic dominance order to the context of random menus, by ordering random compact sets in X through their expected *indirect utilities* for every increasing (and continuous) utility function. As the distribution of a random set is not additive, but is instead a capacity, this requires using the Choquet integral to compute those expectations. This creates a few mathematical complications, but one still gets a fairly basic theory that admits the standard theory of stochastic dominance as a special case.

In Section 2, we give a quick overview of a few basic notions from the theory of random sets and (topological) preordered sets. (This is only meant to be a glossary to streamline the rest of the exposition.) In particular, we introduce a fairly rich class of ordered topological spaces; our main results work so long as X is a member of this class. This framework covers essentially all ordered spaces commonly used in economics, including Euclidean commodity spaces, Banach lattices, lottery spaces ordered by stochastic dominance, etc.. In Section 3, we formally introduce our stochastic ordering $\succsim^{\mathrm{SD}}$ of random menus, and establish several characterizations of it. Most notably, we show that $\succsim^{\mathrm{SD}}$ can be described in terms of probabilities assigned to (the random sets intersecting) up-sets in X, thereby extending the classical characterization of stochastic dominance by distribution functions. In this section we also provide quite a few examples that illustrate the behavior of $\succsim^{\mathrm{SD}}$. Perhaps a bit surprisingly, $\succsim^{\mathrm{SD}}$ exhibits structural features that differ from ordinary stochastic dominance. For instance, it cannot be given a Strassen-type coupling characterization, regardless of how one may choose to compare the realizations of two menus. On a related note, $\succsim^{\mathrm{SD}}$ is more complete than it may at first appear.

Section 4 uses $\succsim^{\mathrm{SD}}$ for making various types of monotone comparative statics for the choice behavior of decision-makers. We first examine when a random choice correspondence that arises from the maximization (of a possibly incomplete) preference relation improves relative to $\succsim^{\mathrm{SD}}$, when either the underlying random menu improves (again relative to $\succsim^{\mathrm{SD}}$) or when the underlying preferences improve. This yields, among other things, stochastic comparative statics results for random maximal sets. We then look at a parametric optimization program where the random choice correspondence is obtained by maximizing a given (deterministic) utility function on a random menu. By means of a Topkis-Milgrom-Shannon type analysis, we find sufficient conditions for this correspondence to be increasing (relative to $\succsim^{\mathrm{SD}}$).

Finally, as a concrete illustration, we deduce stochastic versions of the classical laws of input demand and supply.

Section 5 compares $\succsim^{\mathrm{SD}}$ to another plausible stochastic order, and examines its inheritance behavior under monotone transformations, products, unions, random Minkowski sums, and weak convergence. Almost all of our findings here are positive in that they parallel those that are valid in the case of the stochastic ordering of random variables. Section 6 concludes, and the Appendix reports the proofs of the main results of the paper.

2 Preliminaries

For the convenience of the reader, we collect below the formal definitions of some of the concepts from the theory of capacities and ordered sets. No new results are reported in this section.

Borel Capacities. Let X be a Hausdorff topological space. In what follows, we denote the families of all closed and compact subsets of X by $c(X)$ and $k(X)$, respectively. Besides, $\mathcal{B}(X)$ stands for the Borel σ-algebra on X. Obviously, $k(X) \subseteq c(X) \subseteq \mathcal{B}(X)$. We denote the set of all Borel probability measures on X by $\Delta(X)$.

By a **Borel capacity** on X, we mean an $\supseteq$-increasing function $v : \mathcal{B}(X) \to [0,1]$ with $v(\varnothing) = 0$ and $v(\Omega) = 1$.[2] We say that v is **submodular** if

$$v(A \cup B) + v(A \cap B) \leq v(A) + v(B)$$

for every $A, B \in \mathcal{B}(X)$. If this inequality is reversed, v is said to be **supermodular**. Besides, for any Borel subset S of X, we say that v is **continuous at** S if for any sequences (A_m) and (B_m) in $\mathcal{B}(X)$ with $A_m \uparrow S$ and $B_m \downarrow S$, we have $\lim v(A_m) = v(S) = \lim v(B_m)$.

The Choquet Integral. Let X be a Polish space.[3] For any Borel capacity v on X, and any Borel-measurable map $\varphi : X \to \mathbb{R}$, the **Choquet integral** of φ against v is defined as

$$\mathrm{C}\int_X \varphi \, \mathrm{d}v := \int_0^\infty v\{\varphi \geq t\} \, \mathrm{d}t + \int_{-\infty}^0 (v\{\varphi \geq t\} - 1) \, \mathrm{d}t.$$

(In what follows, we will work with this definition only indirectly.) It is well-known that the map $\varphi \mapsto \mathrm{C}\int_X \varphi \, \mathrm{d}v$ is positively homogeneous, translation invariant, monotonic, and Lipschitz.

[2] See Marinacci and Montrucchio (2004) for a stellar survey of the general theory of capacities and the Choquet integral.

[3] A *Polish space* is a topological space that is homeomorphic to a separable and complete metric space.

Measurable Correspondences. Let (Ω, Σ) be a complete measurable space, and X a Polish space.[4] By a **correspondence** F from Ω to X, we mean any map $F : \Omega \to 2^X$ with $F(\omega) \neq \varnothing$ for all $\omega \in \Omega$; in this case we write $F : \Omega \rightrightarrows X$. Such a correspondence is said to be **closed-valued** if $F(\omega) \in c(X)$ for every $\omega \in X$, and **compact-valued** if $F(\omega) \in k(X)$ for every $\omega \in X$. The **graph** of F is defined as

$$\mathrm{Gr}(F) := \{(\omega, x) \in \Omega \times X : x \in F(\omega)\}.$$

For any $F : \Omega \rightrightarrows X$, the **lower inverse image** of a subset S of X under F is defined as the event that "F hits S." We denote this event by $F_{-1}(S)$, that is,

$$F_{-1}(S) := \{\omega \in \Omega : F(\omega) \cap S \neq \varnothing\}.$$

In turn, F is said to be Σ**-measurable**, or **Effros measurable**, if $F_{-1}(O) \in \Sigma$ for every open subset O of X. As F is compact-valued, and (Ω, Σ) is a complete measurable space, F is Σ-measurable iff any one of the following conditions hold: (i) $F_{-1}(S) \in \Sigma$ for every $S \in c(X)$; (ii) $F_{-1}(S) \in \Sigma$ for every $S \in \mathcal{B}(X)$; and (iii) $\mathrm{Gr}(F)$ is $\Sigma \otimes \mathcal{B}(X)$-measurable.[5]

Notation. Given any probability space $(\Omega, \Sigma, \mathbb{P})$, we denote the set of all compact-valued and Σ-measurable correspondences on Ω by $\mathcal{K}(\mathbb{P}, X)$.

Measurable Selections. Let X be a Polish space, and $F \in \mathcal{K}(\mathbb{P}, X)$. By a Σ**-measurable selection** from F, we mean a Σ-measurable $f : \Omega \to X$ such that $f(\omega) \in F(\omega)$ for $\mathbb{P}$-almost all $\omega \in \Omega$. We denote the set of all such random variables as $\mathrm{Sel}(F)$. The *Kuratowski-Ryll Nardzewski selection theorem* says that there exists a Σ-measurable $f : \Omega \to X$ with $f(\omega) \in F(\omega)$ for all $\omega \in \Omega$, whence $\mathrm{Sel}(F) \neq \varnothing$, for every X-valued Σ-measurable correspondence $F : \Omega \rightrightarrows X$. Moreover, by *Castaing's theorem*, for every $F \in \mathcal{K}(\mathbb{P}, X)$, there exists a sequence (f_m) in $\mathrm{Sel}(F)$ such that $F(\omega) = \mathrm{cl}\{f_1(\omega), f_2(\omega), ...\}$ for every $\omega \in \Omega$.

Preordered Sets. A **preorder** on a nonempty set X is a reflexive and transitive binary relation on X, while an antisymmetric preorder is called a **partial order**. We refer to the pair $(X, \succsim)$ as a **preordered set** (resp., **poset**) if $\succsim$ is a preorder (resp, partial order) on X. (In this case, X is called the **carrier** of $(X, \succsim)$.) For any $x \in X$, we define $x^{\uparrow} := \{y \in X : y \succsim x\}$ and $x^{\downarrow} := \{y \in X : x \succsim y\}$. (In order theory, one refers to these sets as a *principal filter* and a *principal ideal*, respectively.) For any set $A \subseteq X$, we define the **upper closure** of A as $A^{\uparrow} := \bigcup_{x \in A} x^{\uparrow}$, and say that A is an **up-set** if $A = A^{\uparrow}$. (That is, A is an up-set if $y \succsim x \in A$ implies $y \in A$.) Dually, we define $A^{\downarrow} := \bigcup_{x \in A} x^{\downarrow}$, and say that A is a **down-set** if $A = A^{\downarrow}$.

[4] (Ω, Σ) is said to be *complete* if there exists a complete σ-finite measure on Σ.

[5] This is why we work only with *complete* probability spaces in this paper. Without this hypothesis, we could only conclude that $F_{-1}(S)$ is analytic (= Souslin) in Ω for every $S \in \mathcal{B}(X)$.

Given two preordered sets $X = (X, \succsim_X)$ and $Y = (Y, \succsim_Y)$, we say that a map $\varphi : X \to Y$ is **order-preserving** if $x \succsim_X y$ implies $\varphi(x) \succsim_Y \varphi(y)$ for every $x, y \in X$. If $Y = (\mathbb{R}, \geq)$ here, where $\geq$ is the usual ordering of real numbers, we instead refer to φ as $\succsim_X$**-increasing**.

Properly Ordered Spaces. Let $(X, \succsim)$ be a preordered set such that X is a Hausdorff topological space. Following Nachbin (1965), we say that $(X, \succsim)$ is a **normally ordered space** if for every disjoint closed sets A and B in X, with A being an up-set and B a down-set, there exist disjoint open sets O and U in X with $A \subseteq O$ and $B \subseteq U$, and with O being an up-set and U a down-set. If, in addition, $\succsim$ is a closed preorder (i.e, it is a closed subset of $X \times X$), we say $(X, \succsim)$ is a **properly ordered space** . Finally, by a **properly ordered Polish space**, we mean a properly ordered space whose carrier is a Polish space.

The class of properly ordered spaces is broad enough to include essentially all ordered spaces used in economics. For instance, endowing any compact Hausdorff space with a closed preorder yields a properly ordered space. Other examples are given below.

Preordered Banach Spaces. A **preordered Banach space** is a pair $(X, \succsim)$ where X is a Banach space and $\succsim$ is a closed preorder on X such that $x \succsim y$ iff $\lambda x + z \succsim \lambda y + z$ for all $z \in X$ and $\lambda > 0$. (This is equivalent to saying that X is a Banach space with a closed convex cone C such that $x \succsim y$ iff $x - y \in C$.) Any such space is properly ordered.[6]

In particular, every Banach lattice, or more generally, a nonempty closed subset of any Banach lattice, is a properly ordered space. As the class of Banach lattices is quite rich – this class includes all classical Banach spaces with their pointwise orders – this speaks to the generality of the class of properly ordered spaces. In particular, any nonempty closed subset of $\mathbb{R}^n$, ℓ^p (for $1 \leq p < \infty$), $L^p[0,1]$ (for $1 \leq p < \infty$) and $C[0,1]$ are properly ordered Polish spaces (relative to their usual orders).

Notation. Given any properly ordered space $(X, \succsim)$, we denote the set of all continuous real maps on X by $C(X)$. In turn,

$$C_{\uparrow}(X) := \{\varphi \in C(X) : \varphi \text{ is } \succsim \text{-increasing}\}$$

[6] *Proof.* Take any disjoint closed sets A and B in X, with A being an up-set and B a down-set. Define $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ by $f(x) := \text{dist}(x, A)$ and $g(x) := \text{dist}(x, B)$. These are continuous maps with $f^{-1}(0) = A$ and $g^{-1}(0) = B$. Moreover, f is $\succsim$-decreasing and g is $\succsim$-increasing. (Suppose $x \succsim y$, and take any $\varepsilon > 0$. By definition, there is an $a \in A$ with $\|y - a\| < f(y) + \varepsilon$. As $a + (x - y) \in A$ (because A is an up-set), we have $f(x) \leq \|x - a - (x - y)\| = \|y - a\| < f(y) + \varepsilon$. In view of the arbitrariness of ε, then, $f(x) \leq f(y)$. The argument for g is the same.) Our claim is thus proved by choosing $O := \{z \in X : f(z) < g(z)\}$ and $U := \{z \in X : f(z) > g(z)\}$. (As a technical aside, we note that, in view of the Birkhoff-Kakutani metrization theorem, the same proof shows that every first-countable Hausdorff topological linear space that is partially ordered by a closed convex cone is properly ordered.)

and

$$C_{b,\uparrow}(X) := \{\varphi \in C(X) : \varphi \text{ is bounded and } \succsim\text{-increasing}\}.$$

Finally, $L(X)$ stands for the set of all Borel measurable real maps on X. The sets $L_\uparrow(X)$ and $L_{b,\uparrow}(X)$ are defined in the analogous way.

For any X-valued random variable x on $(\Omega, \Sigma, \mathbb{P})$, and bounded $\varphi \in L(X)$, we denote the expectation of $\varphi \circ x$ by $\mathbb{E}(\varphi \circ x)$, that is,

$$\mathbb{E}(\varphi \circ x) := \int_\Omega \varphi \circ x \, \mathrm{d}\mathbb{P}.$$

3 Stochastic Ordering of Random Sets

Throughout this section, $(\Omega, \Sigma, \mathbb{P})$ stands for a complete probability space, and $X = (X, \succsim)$ a properly ordered Polish space.

3.1 Random Sets

Intuitively, a *random set* is a random variable on a probability space whose values are subsets of a given space. One often takes the target space to possess a suitable topological, or linear (or in our case, order-theoretic) structure, and puts some conditions on the types of values that a random set may take (such as closedness, compactness, convexity, etc.). In particular, we will always work with compact-valued random sets in a Polish space.

Formally, by an **X-valued random set** on $(\Omega, \Sigma, \mathbb{P})$, we mean any compact-valued and Σ-measurable correspondence from Ω to X. Thus, the family of all X-valued random sets on $(\Omega, \Sigma, \mathbb{P})$ is precisely $\mathcal{K}(\mathbb{P}, X)$.[7] Every single-valued member of $\mathcal{K}(\mathbb{P}, X)$ can obviously be considered as an X-valued random variable.

Take any $F \in \mathcal{K}(\mathbb{P}, X)$. As it is Effros measurable, stochastic realizations of F can be studied through the probabilities of events of the form "F hits S," as S varies over all Borel subsets of X. Thus, the **distribution** of F is defined as the map $v_F : \mathcal{B}(X) \to [0, 1]$ with

$$v_F(S) := \mathbb{P}\{F \cap S \neq \varnothing\},$$

that is, $v_F(S) = \mathbb{P}(F_{-1}(S))$. When F is single-valued, say, $F = \{x\}$, v_F is, of course, a Borel probability measure on X. This is the *push-forward measure* induced by the X-valued random variable x, and it is denoted by $\mathbb{P}_x$. However, in general, v_F is not additive. All we can say is that it is a submodular capacity that is continuous at $\varnothing$ and X. (It is at this point that the theory of random sets comes to its own.) Some authors refer to v_F as the **capacity induced by** F.

[7] For a comprehensive, and truly excellent, treatment of the theory of closed-valued random sets, see Molchanov (2017). Molchanov and Molinari (2018) is also a brilliant text that introduces the theory of random sets in a way immediately useful for econometrics.

REMARK. One can always think of a random set as a certain type of set-valued random variable. Formally, we can identify any random set $F \in \mathcal{K}(\mathbb{P}, X)$ with the map $\hat{F} : \Omega \to k(X)$ defined by $\hat{F}(\omega) := F(\omega)$. It is well-known that Effros measurability of F implies that $\hat{F}$ is a $k(X)$-valued random variable on $(\Omega, \Sigma, \mathbb{P})$.[8] We denote the pushforward measure induced by $\hat{F}$ on $k(X)$ by $\mathbb{P}_F$. Thus, $\mathbb{P}_F$ is the Borel probability measure on $k(X)$ with $\mathbb{P}_F(\Lambda) := \mathbb{P}\{\omega \in \Omega : F(\omega) \in \Lambda\}$ for any Borel subset Λ of $k(X)$.

Identifying F with $\hat{F}$ allows one adopt the standard machinery developed for random variables to random sets, and we will indeed exploit this below. As we shall see, however, there are major advantages to working with F directly.

3.2 Random Menus

Random sets arise in economics in various garbs. (See Molchanov and Molinari (2018) for numerous examples.) In this paper, we typically think of a random set as a feasible set from which a choice will be made, once that set is realized. This interpretation aligns naturally with the decision-theoretic literature on menu choice. For instance, take the context of consumer choice where $\mathbb{R}^n_+$ models the space of all commodity bundles (of n goods). Let p and w be $\mathbb{R}^n_{++}$- and $\mathbb{R}_+$-valued random variables on $(\Omega, \Sigma, \mathbb{P})$, modeling the stochastic prices and income, respectively. Then, the associated *random budget set* is $b \in \mathcal{K}(\mathbb{P}, \mathbb{R}^n_+)$ with

$$b(\omega) := \{x \in \mathbb{R}^n_+ : p_1(\omega)x_1 + \cdots + p_n(\omega)x_n \leq w(\omega)\}.$$

We will consider several other examples in Sections 3.6 and 4.

In what follows, when we interpret a random set in this manner, we will often call it a *random menu* . Formally, there is no difference between terms "random set" and "random menu," but the latter term has the advantage of highlighting the motivation behind the stochastic ordering notion we advance next.

3.3 Stochastic Dominance of Random Menus

For any random set $F \in \mathcal{K}(\mathbb{P}, X)$ and a bounded and Borel measurable $\varphi : X \to \mathbb{R}$, we define the function $s_{F,\varphi} : \Omega \to \mathbb{R}$ by

$$s_{F,\varphi}(\omega) := \sup_{z \in F(\omega)} \varphi(z).$$

Using Castaing's theorem, one can easily show that $s_{F,\varphi}$ is Σ-measurable. Furthermore,

$$\int_\Omega s_{F,\varphi} \, \mathrm{d}\mathbb{P} = \mathrm{C} \int_X \varphi \, \mathrm{d}v_F. \tag{1}$$

[8]Here we think of $k(X)$ as topologized by the Hausdorff metric induced by any metric on X that metrizes the topology of X. (As X is Polish, this is equivalent to endowing $k(X)$ with the Vietoris topology.) Then, by $\hat{F}$ being a $k(X)$-valued random variable, we mean that it is $\Sigma \backslash \mathcal{B}(k(X))$-measurable.

This is given as Theorem 1.6.3 in Molchanov (2017) for $[0,\infty)$-valued φ, but, as noted in that text as well, it is easily extended to the general case.

Intuitively, we can think of $s_{F,\varphi}$ as the (random) indirect utility function of a person with Bernoulli utility φ and constraint set F. (See Figure 1.) In turn, $\int_\Omega s_{F,\varphi}\,\mathrm{d}\mathbb{P}$ is the expected indirect utility of this person. (1) says that we can conveniently represent this expectation as the Choquet integral of φ against the distribution of F. Of course, if F is single-valued, say, $F=\{x\}$ for some X-valued random variable x on $(\Omega,\Sigma,\mathbb{P})$, this expectation is simply the von Neumann-Morgenstern utility of x (relative to φ).

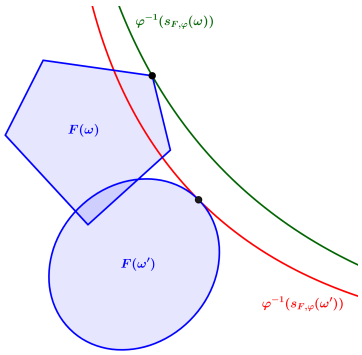


Figure 1: The green and red lines are the highest level curves of φ touching $F(\omega)$ and $F(\omega')$, respectively.

There are many equivalent ways in which one can define a stochastic ordering for X-valued random variables x and y on $(\Omega,\Sigma,\mathbb{P})$. The one that is intuitively most appealing is the following: x (*first-order*) **stochastically dominates** y – we denote this by writing $x \succsim_{\mathrm{FSD}} y$ – if $\mathbb{E}(\varphi\circ x)\geq\mathbb{E}(\varphi\circ y)$ for every $\succsim$-increasing, continuous and bounded real map φ on X, that is, for every $\varphi\in C_{b,\uparrow}(X)$. Thus, $x\succsim_{\mathrm{FSD}} y$ holds whenever the expected utility of x is higher than that of y for every person whose (Bernoulli) utility function on X is increasing (relative to the order of X). The preorder $\succsim_{\mathrm{FSD}}$ generalizes the coverage of stochastic dominance theory substantially; it was introduced in Kamae, Krengel and O'Brien (1977). When $X=\mathbb{R}$, it obviously reduces to the (first-order) stochastic dominance that is widely used in economics and finance, among other fields.

We extend $\succsim_{\mathrm{FSD}}$ to the context of X-valued random menus by replacing the role of "expected utilities" with "expected indirect utilities." Formally, we define the (*first-order*) **stochastic dominance order** on $\mathcal{K}(\mathbb{P},X)$ as the binary relation $\succsim^{\mathrm{SD}}$ on this set with

$$F\succsim^{\mathrm{SD}} G \quad \text{iff} \quad \int_\Omega s_{F,\varphi}\,\mathrm{d}\mathbb{P}\geq\int_\Omega s_{G,\varphi}\,\mathrm{d}\mathbb{P} \text{ for every } \varphi\in C_{b,\uparrow}(X).$$

Obviously, $\succsim^{\mathrm{SD}}$ is a preorder on $\mathcal{K}(\mathbb{P},X)$ that is consistent with $\succsim_{\mathrm{FSD}}$ in the case of X-valued random variables on $(\Omega,\Sigma,\mathbb{P})$.

In view of (1), the stochastic dominance order for random menus can be expressed in terms of direct utilities as well, provided that we define "expected utility" by using the Choquet integral against the distribution of the given random set. Put precisely: For any $F, G \in \mathcal{K}(\mathbb{P}, X)$,

$$F \succsim^{\mathrm{SD}} G \quad \text{iff} \quad \mathrm{C}\int_X \varphi \,\mathrm{d}v_F \geq \mathrm{C}\int_X \varphi \,\mathrm{d}v_G \text{ for every } \varphi \in C_{b,\uparrow}(X).$$

This perhaps better witnesses how $\succsim^{\mathrm{SD}}$ extends the first order stochastic dominance order from the realm of random variables to that of random sets.

Another way of looking at the statement $F \succsim^{\mathrm{SD}} G$ is the following. Any one selection f from the random menu F can be viewed as a contingent plan of action (to be implemented when the uncertainty resolves). Then, for any Bernoulli utility φ on X, $\mathbb{E}(\varphi \circ f)$ is the expected utility of the plan f. The "value" of F relative to φ may thus be thought of as the maximum of these expected utilities across all $f \in \mathrm{Sel}(F)$. It then seems quite meaningful to say that F *dominates* G iff the "value" of the menu F exceeds that of G regardless of which $\succsim$-increasing Bernoulli utility φ we use in the comparison. In fact, as we show next, $\succsim^{\mathrm{SD}}$ compares random menus precisely in this manner.

PROPOSITION 1. *Let $(\Omega, \Sigma, \mathbb{P})$ be a complete probability space, and $(X, \succsim)$ a properly ordered Polish space. For any $F, G \in \mathcal{K}(\mathbb{P}, X)$, we have $F \succsim^{\mathrm{SD}} G$ if, and only if, for every $\varphi \in C_{b,\uparrow}(X)$ there exists an $f_\varphi \in \mathrm{Sel}(F)$ with*

$$\mathbb{E}(\varphi \circ f_\varphi) \geq \max_{g \in \mathrm{Sel}(G)} \mathbb{E}(\varphi \circ g).$$

PROOF. Take any $F, G \in \mathcal{K}(\mathbb{P}, X)$ and $\varphi \in C_{b,\uparrow}(X)$. As F is compact-valued, $s_{F,\varphi}(\omega) = \max_{z \in F(\omega)} \varphi(z)$ for every $\omega \in \Omega$. By the measurable maximum theorem (Aliprantis and Border (2006), Theorem 18.19), then, there is an $f_\varphi \in \mathrm{Sel}(F)$ with $\varphi \circ f_\varphi = s_{F,\varphi}$. Obviously, $\mathbb{E}(\varphi \circ f_\varphi) \geq \mathbb{E}(\varphi \circ f)$ for every $f \in \mathrm{Sel}(F)$, so we have $\mathbb{E}(\varphi \circ f_\varphi) = \max_{f \in \mathrm{Sel}(F)} \mathbb{E}(\varphi \circ f)$, and of course the same fomula holds in the case of G as well. Our claim follows readily from these formulas and the definition of $\succsim^{\mathrm{SD}}$. ■

By definition, $\succsim^{\mathrm{SD}}$ compares random menus through their expected indirect utilities. Proposition 1 shows that this comparison admits an equivalent formulation in terms of the maximal expected utilities that can be attained from the menus by measurable contingent choices. This proposition is also quite serviceable, as we illustrate next.

EXAMPLE 1. It is straightforward to see how $\succsim^{\mathrm{SD}}$ compares a random set with a random singleton (= random variable). Take any $F, G \in \mathcal{K}(\mathbb{P}, X)$ and any X-valued random variables x and y on $(\Omega, \Sigma, \mathbb{P})$. Proposition 1 entails that $F \succsim^{\mathrm{SD}} y$ iff for

every $\varphi \in C_{b,\uparrow}(X)$ there exists an $f_\varphi \in \mathrm{Sel}(F)$ with $\mathbb{E}(\varphi \circ f_\varphi) \geq \mathbb{E}(\varphi \circ y)$. (Here we denote the map $\omega \mapsto \{y(\omega)\}$ simply by y.) In other words,

$$F \succsim^{\mathrm{SD}} y \quad \text{iff} \quad \inf \sup_{f \in \mathrm{Sel}(F)} (\mathbb{E}(\varphi \circ f) - \mathbb{E}(\varphi \circ y)) \geq 0$$

where the infimum is taken over all $\varphi \in C_{b,\uparrow}(X)$. On the other hand, again by Proposition 1, $x \succsim^{\mathrm{SD}} G$ iff $x \succsim_{\mathrm{FSD}} g$ for every $g \in \mathrm{Sel}(G)$.

Relation to the Literature. Molchanov (2017), Section 1.5.5, introduces a stochastic dominance ordering for random sets by identifying F with the $k(X)$-valued random variable $\hat{F}$ as in the Remark at the end of Section 3.1. In the case where X is locally compact, Molchanov's preorder $\succsim$ is defined on $\mathcal{K}(\mathbb{P}, X)$ as $F \succsim G$ iff $\hat{F} \succsim_{\mathrm{FSD}} \hat{G}$ where $k(X)$ is ordered by the containment ordering (and topologized by the Fell topology). It is easy to show that this preorder is stronger than $\succsim^{\mathrm{SD}}$. The converse does not hold. Indeed, Molchanov's preorder is quite demanding. If F and G are single-valued, then they are comparable by $\succsim$ iff F and G are equal in distribution. Thus, unlike $\succsim^{\mathrm{SD}}$, Molchanov's preorder does not reduce to the standard stochastic dominance order in the case of single-valued random sets. (This is one instance where the identification of F and $\hat{F}$ does not lead to a desirable conclusion.)

The only other stochastic orders for random sets that we are aware of are the ones introduced in Fernández and Molchanov (2003) and Cerreia-Vioglio and Ok (2025). These orders are, however, patterned after the convex order of random variables, and target measuring the variability of random sets (as opposed to their desirability). As such, they are unrelated to $\succsim^{\mathrm{SD}}$.

3.4 The Characterization Theorem

One of the most useful facts in the classical theory of stochastic orders is that stochastic dominance can be characterized in terms of probabilities assigned to up-sets.[9] Such characterizations let one verify stochastic comparisons directly from the distributions of the involved random variables. Our next result shows that an analogous characterization is valid in the present context as well, provided we use the capacities induced by random menus. As we will demonstrate subsequently, this makes the theory more amenable to applications.

THEOREM 2. *Let $(\Omega, \Sigma, \mathbb{P})$ be a complete probability space, and $(X, \succsim)$ a properly ordered Polish space. For any random sets $F, G \in \mathcal{K}(\mathbb{P}, X)$, the following are equivalent:*

i. $F \succsim^{\mathrm{SD}} G$;

ii. $v_F(S) \geq v_G(S)$ for every closed up-set S in X;

iii. $v_F(S) \geq v_G(S)$ for every Borel up-set S in X;

[9] For a quick, but rigorous, introduction to the theory of stochastic orders, see Marshall (1991), and Mosler and Scarsini (1991).

iv. $\mathbb{E}(s_{F,\varphi}) \geq \mathbb{E}(s_{G,\varphi})$ *for every* $\varphi \in L_{b,\uparrow}(X)$.

If we apply Theorem 2 to the case of single-valued random sets, we recover the classical characterization of Kamae, Krengel and O'Brien (1977), this time in the more general context of properly ordered Polish spaces. In particular, in the context of Theorem 2, for any two X-valued random variables x and y on $(\Omega, \Sigma, \mathbb{P})$, we have $\mathbb{E}(\varphi \circ x) \geq \mathbb{E}(\varphi \circ y)$ for every $\varphi \in C_{b,\uparrow}(X)$ if, and only if, $\mathbb{P}\{x \in S\} \geq \mathbb{P}\{y \in S\}$ for every Borel (or for every closed) up-set S in X. Obviously, when $X = \mathbb{R}$, this reduces to the well-known characterization of the stochastic dominance ordering in terms of distribution functions.

In passing, we reiterate that Theorem 2 is quite general in that it applies to the case of all random sets that take their values in any separable Banach lattice (such as $\mathbb{R}^n$, or p-summable sequence spaces, or $C[0,1]$, etc.), among others.

3.5 Set-Valued Transformations of Random Variables

Many economically relevant random menus are obtained by applying a particular set-valued transformation to an underlying random variable. In such environments, one would expect ordinary stochastic improvements in the underlying uncertainty (in terms of that random variable) to induce stochastic improvements in the associated random menu, provided the transformation is suitably monotonic. We will show next that $\succsim^{\mathrm{SD}}$ obeys this intuition.

Let $X = (X, \succsim_X)$ and $Y = (Y, \succsim_Y)$ be two properly ordered Polish spaces, and $\Gamma : X \rightrightarrows Y$ a correspondence. There are multiple ways in which we can think of Γ as *monotonic* . As it is particularly suitable for our purposes, we say that Γ is (*upper-*) **increasing** if $x \succsim_X y$ implies that for every $b \in \Gamma(y)$ there is an $a \in \Gamma(x)$ with $a \succsim_Y b$.

For any Y-valued random variable z on $(\Omega, \Sigma, \mathbb{P})$ and $\Gamma : X \rightrightarrows Y$, the correspondence $\Gamma \circ z$ is a random set on $(\Omega, \Sigma, \mathbb{P})$, provided that $\Gamma_{-1}(S)$ is Borel for every closed $S \subseteq X$. (Of course, we define $\Gamma \circ z : \Omega \rightrightarrows X$ by $\Gamma \circ z(\omega) := \Gamma(z(\omega))$.) When Γ is increasing, and there is a stochastic comparison between two random variables x and y, this comparison is inherited by $\Gamma \circ x$ and $\Gamma \circ y$. The following corollary of Theorem 2 demonstrates this formally. We will make repeated use of this fact in the next section.

PROPOSITION 3. *Let* $(\Omega, \Sigma, \mathbb{P})$ *be a complete probability space, and* $(X, \succsim_X)$ *and* $(Y, \succsim_Y)$ *properly preordered Polish spaces. Let* $\Gamma : X \rightrightarrows Y$ *be a compact-valued correspondence such that* $\Gamma_{-1}(S) \in \mathcal{B}(X)$ *for every closed* $S \subseteq X$. *Then,* Γ *is increasing if, and only if, for any* X*-valued random variables* x *and* y *on* $(\Omega, \Sigma, \mathbb{P})$,

$$x \succsim_{\mathrm{FSD}} y \quad \textit{implies} \quad \Gamma \circ x \succsim^{\mathrm{SD}} \Gamma \circ y.$$

PROOF. Applying Theorem 2 to the set-valued maps $\omega \mapsto \{x(\omega)\}$ and $\omega \mapsto \{y(\omega)\}$ shows that $x \succsim_{\mathrm{FSD}} y$ is equivalent to say $\mathbb{P}\{x \in A\} \geq \mathbb{P}\{y \in A\}$ for every Borel

up-set A in X. Now suppose Γ is increasing, and take any closed up-set S in Y. Then, $\Gamma_{-1}(S)$ is an up-set in X. (Indeed, suppose $a \succsim_X b \in \Gamma_{-1}(S)$. By definition of $\Gamma_{-1}(S)$, there exists some t in $S \cap \Gamma(b)$. So, as Γ is increasing, $a \succsim_X b$ implies there is an $s \in \Gamma(a)$ with $s \succsim_Y t$. Since S is an up-set in Y and $t \in S$, this yields $s \in S$. But then $S \cap \Gamma(a) \neq \varnothing$, that is, $a \in \Gamma_{-1}(S)$.) Moreover, $\Gamma_{-1}(S)$ is a Borel subset of X, because Γ is $\mathcal{B}(X)$-measurable. It follows that $\mathbb{P}\{x \in \Gamma_{-1}(S)\} \geq \mathbb{P}\{y \in \Gamma_{-1}(S)\}$, that is, $\mathbb{P}\{(\Gamma \circ x) \cap S \neq \varnothing\} \geq \mathbb{P}\{(\Gamma \circ y) \cap S \neq \varnothing\}$. In view of the arbitrary choice of S and Theorem 2, we conclude that $\Gamma \circ x \succsim^{\mathrm{SD}} \Gamma \circ y$. Conversely, suppose Γ is not increasing. This means there exist $b \in Y$ and $\alpha, \beta \in X$ such that $\alpha \succsim_X \beta$ and $b \in \Gamma(\beta)$ but $a \succsim_Y b$ fails for all $a \in \Gamma(\alpha)$. Then, where we denote the random variable that equals α everywhere again by α (and similarly for β), $\alpha \succsim_{\mathrm{FSD}} \beta$, and yet $\Gamma(\beta) \cap b^{\uparrow} \neq \varnothing = \Gamma(\alpha) \cap b^{\uparrow}$, so $\Gamma \circ \alpha \succsim^{\mathrm{SD}} \Gamma \circ \beta$ fails. $\square$

3.6 Examples

We begin with a very simple example that nevertheless illustrates the nontrivial nature of $\succsim^{\mathrm{SD}}$.

EXAMPLE 2. Let $X = \{a, b, c\}$, and order this set with the equality relation. Let us consider the X-valued random sets F and G on $(\Omega, \Sigma, \mathbb{P})$ such that F equals $\{a, b\}$ and $\{a, c\}$, each with probability $\frac{1}{2}$, and G equals X and $\{a\}$, each with probability $\frac{1}{2}$. (See Figure 2.) It may not seem obvious at the outset if one can compare the desirability of these random menus in some objective way. But, evidently, $v_F \geq v_G$ (and $v_F\{b, c\} = 1 > \frac{1}{2} = v_G\{b, c\}$). So, by Theorem 2, $F \succsim^{\mathrm{SD}} G$.



Figure 2: Illustration of Example 2.

REMARK. Let F and G be X-valued random sets on $(\Omega, \Sigma, \mathbb{P})$. By a **coupling** for (F, G), we mean a pair (F', G') of X-valued random sets on $(\Omega, \Sigma, \mathbb{P})$ such that $F = F'$ and $G = G'$, both in distribution. Notice that in the example above there is no coupling (F', G') for (F, G) such that $F'(\omega) \supseteq G'(\omega)$ for $\mathbb{P}$-almost all $\omega \in \Omega$. Thus, $\succsim^{\mathrm{SD}}$ does not have a Strassen type characterization relative to $\supseteq$. (In particular, $\succsim^{\mathrm{SD}}$ differs from Molcahnov's order even in the context of this simple example.)

The culprit here is actually not the incompleteness of the containment ordering. In the case where $X = \{a, b, c\}$, one can show that for *any* preorder $\trianglerighteq$ on 2^X, there

exist X-valued random sets F and G on some $(\Omega, \Sigma, \mathbb{P})$ (each taking two values with equal probabilities) such that $F \gtrsim^{\mathrm{SD}} G$, but there is no coupling (F', G') for (F, G) with $F' \trianglerighteq G'$ almost surely. (The latter condition is equivalent to saying that $\hat{F} \gtrsim_{\mathrm{FSD}} \hat{G}$ fails, where the discrete space 2^X is ordered by $\trianglerighteq$.)[10] This suggests that $\gtrsim^{\mathrm{SD}}$ is indeed structurally different than ordinary stochastic dominance.

Our next set of examples are geometric in nature, and provides further instances of the nontrivial behavior of $\gtrsim^{\mathrm{SD}}$.

EXAMPLE 3. (*Random Intervals*) Let x, y, x' and y' be four (real-valued) random variables on a probability space such that $x \leq y$ and $x' \leq y'$. Then, by Theorem 2, the random interval $[x, y]$ first-order stochastically dominates $[x', y']$ iff probability of the event $\{a \leq y\}$ exceeds that of $\{a \leq y'\}$ for every $a \in \mathbb{R}$. Conclusion: $[x, y] \gtrsim^{\mathrm{SD}} [x', y']$ iff $y \gtrsim_{\mathrm{FSD}} y'$. When we look at a random interval as a random menu, this conclusion is unexceptionable. Conversely, it shows that $\gtrsim^{\mathrm{SD}}$ is not the right ordering to use if one wishes to incorporate the "size" of random sets as a whole.

EXAMPLE 4. (*Random Balls around the Origin*) Take any integer $n \geq 2$, any $p \in [1, \infty]$, and consider $\mathbb{R}^n$ as metrized by the p-norm $\|\cdot\|_p$.[11] For any $r > 0$, we denote the closed r-ball around any $u \in \mathbb{R}^n$ by $B_p[u, r]$. In turn, for any nonnegative random variable z on $(\Omega, \Sigma, \mathbb{P})$, we denote the correspondence that maps any $\omega \in \Omega$ to $B_p[\mathbf{0}, z(\omega)]$ as $B_p[\mathbf{0}, z]$. (Here $\mathbf{0}$ is the n-vector of 0s.) Then, for any such random variables x and y,

$$x \gtrsim_{\mathrm{FSD}} y \quad \text{if and only if} \quad B_p[\mathbf{0}, x] \gtrsim^{\mathrm{SD}} B_p[\mathbf{0}, y].$$

The "only if" part of this claim follows from Proposition 3. Conversely, take any real number $a \geq 0$, and define $w := a(n^{-1/p}, ..., n^{-1/p})$. Then, $\|w\|_p = a$, whence $x \geq a$ iff $B_p[\mathbf{0}, x] \cap w^{\uparrow} \neq \varnothing$, and similarly for y. Thus, by Theorem 2, $B_p[\mathbf{0}, x] \gtrsim^{\mathrm{SD}} B_p[\mathbf{0}, y]$ implies

$$\mathbb{P}\{x \geq a\} = \mathbb{P}\{B[\mathbf{0}, x] \cap w^{\uparrow} \neq \varnothing\} \geq \mathbb{P}\{B[\mathbf{0}, y] \cap w^{\uparrow} \neq \varnothing\} = \mathbb{P}\{y \geq a\}.$$

In view of the arbitrary choice of a, we conclude $x \gtrsim_{\mathrm{FSD}} y$.

EXAMPLE 5. (*Random Balls with Fixed Radius*) We again work within the framework of the previous example. Fix an arbitrary $r > 0$, and for any $\mathbb{R}^n$-valued random variable z on $(\Omega, \Sigma, \mathbb{P})$, denote the correspondence that maps any $\omega \in \Omega$ to $B_p[z(\omega), r]$ by $B_p[z, r]$. In this case too

$$x \gtrsim_{\mathrm{FSD}} y \quad \text{implies} \quad B_p[x, r] \gtrsim^{\mathrm{SD}} B_p[y, r]$$

[10] To wit, if we choose $\trianglerighteq$ to be the cardinality ordering (or any other ordering that extends $\supseteq$), $\hat{F} \gtrsim_{\mathrm{FSD}} \hat{G}$ fails for the random sets F and G of Example 2. (Try, for instance, the test function φ that assigns 0 to singletons, 2 to doubletons, and 3 to X.)

[11] Unless explicitly stated otherwise, we always consider $\mathbb{R}^n$ as partially ordered componentwise.

by Proposition 3. However, the converse implication does not hold when $p < \infty$. (This again shows that $\succsim^{\text{SD}}$ is apparently more complete than one may initially surmise.) To illustrate, suppose $n = 2$ and $r = 1$. For any $\varepsilon > 0$, consider the random 2-vectors x_ε and y, where x_ε takes values $(1, -\varepsilon)$ and $(-\varepsilon, 1)$, each with probability $\frac{1}{2}$, and y takes values $\mathbf{0} := (0, 0)$ and $(-1, -1)$, each with probability $\frac{1}{2}$. Then, for any $p \in [1, \infty)$, we can choose $\varepsilon > 0$ small enough that $B_p[\mathbf{0}, r]$ is contained in $B_p[(1, -\varepsilon), r]^{\downarrow} \cup B_p[(-\varepsilon, 1), r]^{\downarrow}$ – see Figure 3. This entails that if $B_p[\mathbf{0}, r] \cap S \neq \varnothing$ for a closed up-set S in $\mathbb{R}^2$, either $B_p[(1, -\varepsilon), r] \cap S \neq \varnothing$ or $B_p[(-\varepsilon, 1), r] \cap S \neq \varnothing$. Thus, by Theorem 2, $B_p[x, r] \succsim^{\text{SD}} B_p[y, r]$. Yet $x \succsim_{\text{FSD}} y$ is false.

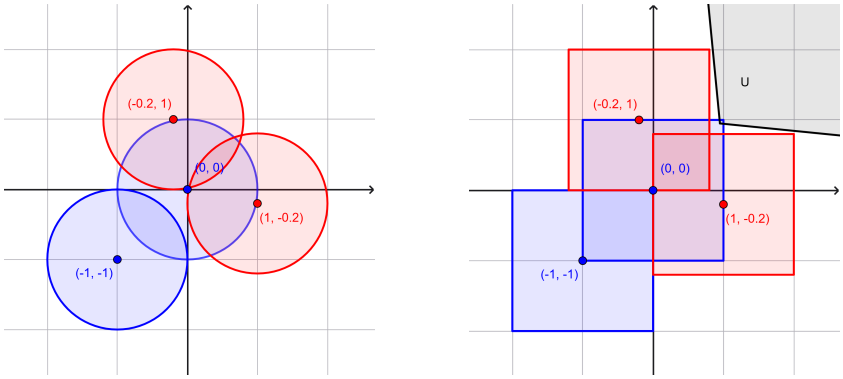


Figure 3: Illustration of Example 5 (left) and 6 (right) with $r = 1$ and $\varepsilon = 0.2$.

EXAMPLE 6. (*Random Cubes with Fixed Radius*) The restriction we imposed on the curvature of the balls (i.e., the hypothesis $p < \infty$) was essential in the previous example. The second part of Figure 3 shows that the type of construction we just presented does not work with the sup-metric. Indeed, in the context of $\mathbb{R}^n$,

$$x \succsim_{\text{FSD}} y \quad \text{if and only if} \quad B_\infty[x, r] \succsim^{\text{SD}} B_\infty[y, r]$$

for any (fixed) $r > 0$. To see this, take any $\varphi \in C_{b,\uparrow}(X)$, write $\mathbf{1}$ for the n-vector of 1s, and notice that $s_{B_\infty[z,r],\varphi} = \varphi(z + r\mathbf{1})$ for any $\mathbb{R}^n$-valued random variable z on $(\Omega, \Sigma, \mathbb{P})$. Thus, $B_\infty[x, r] \succsim^{\text{SD}} B_\infty[y, r]$ iff $\mathbb{E}(\varphi(x + r\mathbf{1})) \geq \mathbb{E}(\varphi(y + r\mathbf{1}))$ for every $\varphi \in C_{b,\uparrow}(X)$. The latter condition says $x + r\mathbf{1} \succsim_{\text{FSD}} y + r\mathbf{1}$, that is, $x \succsim_{\text{FSD}} y$.

We next turn to examples drawn from economics.

EXAMPLE 7. (*Random Budget Sets*) For any integer $n \geq 2$, $\alpha > 0$, and $\beta \in \mathbb{R}^n_{++}$, we define $b(\alpha, \beta) := \{x \in \mathbb{R}^n_+ : \sum^n \beta_i x_i \leq \alpha\}$, the budget set of a consumer with income α at the price vector β. Then, where w and w' are positive random variables on $(\Omega, \Sigma, \mathbb{P})$, we have

$$w \succsim_{\text{FSD}} w' \quad \text{if and only if} \quad b(w, \beta) \succsim^{\text{SD}} b(w', \beta)$$

for any $\beta \in \mathbb{R}^n_{++}$. The "only if" part of this follows from Proposition 3. Conversely, fix any $\beta \in \mathbb{R}^n_{++}$, and suppose $b(w,\beta) \succsim^{\text{SD}} b(w',\beta)$. Now take any $\alpha > 0$, and put $x := (\alpha/\beta_1, 0, ..., 0)$. Then, $\mathbb{P}\{w \geq \alpha\} = \mathbb{P}\{b(w,\beta) \cap x^{\uparrow} \neq \varnothing\} \geq \mathbb{P}\{b(w',\beta) \cap x^{\uparrow} \neq \varnothing\} = \mathbb{P}\{w' \geq \alpha\}$. In view of the arbitrariness of α, we conclude that $w \succsim_{\text{FSD}} w'$.

If we model prices stochastically, the result changes: For any $\mathbb{R}^n_{++}$-valued random variables p and p' on $(\Omega, \Sigma, \mathbb{P})$,

$$p' \succsim_{\text{FSD}} p \quad \text{implies} \quad b(p,\alpha) \succsim^{\text{SD}} b(p',\alpha)$$

for any $\alpha > 0$, *but not conversely*. The first claim again follows from Proposition 3. To prove the second, take $n = 2$ and $\alpha = 12$. Let p take values $(4,2)$ and $(2,4)$, each with probability $\frac{1}{2}$, and p' take values $(6,6)$ and $(3,3)$, each with probability $\frac{1}{2}$. Then, $b(12,(3,3))$ is contained within the union of $b(12,(4,2))$ and $b(12,(2,4))$; see Figure 4. It thus follows from Theorem 2 that $b(12,p) \succsim^{\text{SD}} b(12,p')$, and yet $p' \succsim_{\text{FSD}} p$ is false. (This again witnesses that stochastic dominance for random menus that are defined parametrically goes beyond the stochastic dominance of the involved parameters).

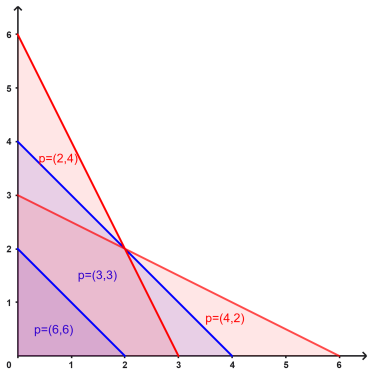


Figure 4: Illustration of Example 7.

EXAMPLE 8. (*Random Core*) Take any $n \in \mathbb{N}$, and put $N := \{1, ..., n\}$. A (transferable utility) *coalitional game* v on N is a capacity on 2^N. (In particular, $v(N) = 1$.) Let V_N denote the set of all coalitional games on N with nonempty cores. Then, by Proposition 3,

$$v' \succsim_{\text{FSD}} v \quad \text{implies} \quad \text{core}(v) \succsim^{\text{SD}} \text{core}(v')$$

for any V_N-valued random variables on $(\Omega, \Sigma, \mathbb{P})$. And again, the converse is false. To wit, let $n = 2$ so that we can represent any capacity v by the 2-vector $(v\{1\}, v\{2\})$. Let v take values $(.2, .4)$ and $(.4, .2)$, each with probability $\frac{1}{2}$, and v' take values $(.5, .5)$ and $(.3, .3)$, each with probability $\frac{1}{2}$. It is easy to check that $\text{core}(v) \succsim^{\text{SD}} \text{core}(v')$ but not $v' \succsim_{\text{FSD}} v$.

EXAMPLE 9. (*Random Maximal Sets*) Let $(X, \succsim)$ be a properly ordered com-

pact Hausdorff space.[12] For any nonempty $A \subseteq X$, we denote the set of all $\succsim$-maximal elements in A by $\textsc{max}(A, \succsim)$, that is, $\textsc{max}(A, \succsim) := \{y \in A : x \succ y \text{ for no } x \in A\}$. It is well-known that this set is nonempty whenever A is nonempty and compact. Thus, for any $F \in \mathcal{K}(\mathbb{P}, X)$, the set-valued map $\omega \mapsto \text{cl}(\textsc{max}(F(\omega), \succsim))$ is a compact-valued correspondence from Ω into X. While this is not obvious, this correspondence is Effros measurable. (See Theorem D in the Appendix.) Consequently, $\Phi : \mathcal{K}(\mathbb{P}, X) \to \mathcal{K}(\mathbb{P}, X)$ is well-defined by the formula

$$\Phi(F)(\omega) := \text{cl}(\textsc{max}(F(\omega), \succsim)), \quad \omega \in \Omega.$$

It turns out that Φ is order-preserving relative to $\succsim^{\text{SD}}$. In fact:

$$F \succsim^{\text{SD}} G \quad \text{if and only if} \quad \Phi(F) \succsim^{\text{SD}} \Phi(G).$$

To see this, take any $\varphi \in C_{\uparrow}(X)$. Obviously, $s_{F,\varphi} \geq s_{\Phi(F),\varphi}$. Conversely, take any $\omega \in \Omega$ and $x \in F(\omega)$. As $F(\omega) \cap x^{\uparrow}$ is compact, it contains a $\succsim$-maximal element, say, y. But then y is $\succsim$-maximal in $F(\omega)$ as well. Thus, $y \succsim x$, whence $\varphi(y) \geq \varphi(x)$, for some $y \in \textsc{max}(F(\omega), \succsim)$. It follows that $s_{F,\varphi} \leq s_{\Phi(F),\varphi}$. Thus, $s_{F,\varphi} = s_{\Phi(F),\varphi}$, and our claim follows from this observation.

4 Application: Monotone Comparative Statics

One can utilize the stochastic order $\succsim^{\text{SD}}$ to make various types of *ex ante* monotone comparative statics. In this section we demonstrate this in terms of the choice behavior of rational decision makers.

4.1 Comparative Choice Across Random Menus

Let X be a Polish space, and $\succsim$ a closed preorder on X. We presently interpret X as a space of mutually exclusive alternatives, and think of $\succsim$ as ordering the "magnitude" of these alternatives in some *objective* way. For instance, if the model is one of consumer choice, we may take X as $\mathbb{R}^n$, and $\succsim$ as the usual order of $\mathbb{R}^n$ that ranks consumption vectors componentwise. Or, if X is a monetary lottery space that consists of probability measures on $\mathbb{R}$, we may take $\succsim$ as the standard first-order stochastic dominance. This would rank the lotteries on the basis of their desirability in an objective way. Or, if the model is one of social choice, where we are given a collection of individuals each with a utility function on X, we may take $\succsim$ as the Pareto ordering.

Now consider a rational person whose preferences on X are represented by means of a closed preorder $\trianglerighteq$ on X (whose strict part we denote by $\triangleright$). Thus, we assume that preferences are transitive, but not necessarily complete. Moreover, as we presently interpret $\succsim$ as an objective ordering of the alternatives, we assume $\succ \subseteq \triangleright$.

[12] If $\succsim$ is a partial order, then we can take X as a properly ordered Polish space in this example.

For instance, in the case of the consumer choice model, this just means the consumer prefers more to less. In the case of lottery choice, it means the preferences of the individial are stochastically monotonic, and in the case of social choice, it just means the social planner respects the Pareto principle.

We model the choices of this individual at any nonempty compact menu $A \subseteq X$ by maximizing $\trianglerighteq$ on A. Thus, the *ex post* **choice correspondence** of this person is $C(\cdot, \trianglerighteq) : k(X) \rightrightarrows X$ where

$$C(A, \trianglerighteq) := \text{MAX}(A, \trianglerighteq).$$

Consequently, to model the situation where the feasible menus are stochastically realized, we define the *ex ante* **choice correspondence** of this person as the map $\mathcal{C}(\cdot, \trianglerighteq) : \mathcal{K}(\mathbb{P}, X) \to \mathcal{K}(\mathbb{P}, X)$ with

$$\mathcal{C}(F, \trianglerighteq)(\omega) := \text{cl}(C(F(\omega), \trianglerighteq)).$$

If X is compact, this map is well-defined (Theorem D in the Appendix). The question we pose is this: If one random menu F stochastically dominates another random menu G (this only uses the preorder $\succsim$ as input), when does it follow that the (random) choice set of the individual on F stochastically dominates that on G? Put formally, we wish to determine the structure of preferences $\trianglerighteq$ so that

$$F \succsim^{\text{SD}} G \quad \text{implies} \quad \mathcal{C}(F, \trianglerighteq) \succsim^{\text{SD}} \mathcal{C}(G, \trianglerighteq) \tag{2}$$

for every $F, G \in \mathcal{K}(\mathbb{P}, X)$. The answer turns out to be quite crisp:

FACT 1. (2) holds for all $F, G \in \mathcal{K}(\mathbb{P}, X)$ only if $\succ = \triangleright$.

PROOF. Suppose $\succ \neq \triangleright$. As $\succ \subseteq \triangleright$ by hypothesis, it follows that $x \triangleright y$ but not $x \succ y$ for some $x, y \in X$. Set $A := \{x, y\}$. Then $y \notin C(A, \trianglerighteq)$ but $y \in C(A, \succsim)$. By the Strict Separation Lemma (Lemma C of the Appendix), there exists a continuous and $\succsim$ -increasing map $\varphi : X \to [0, 1]$ with $\varphi^{-1}(1) = y^{\uparrow}$. (Here $y^{\uparrow}$ is the principal filter relative to $\succsim$.) Now consider the constant random sets F and G with $F = \{x, y\}$ and $G = \{y\}$. Clearly, $F \succsim^{\text{SD}} G$ while $\mathbb{E}(s_{\mathcal{C}(F, \trianglerighteq), \varphi}) = \varphi(x) < \varphi(y) = \mathbb{E}(s_{\mathcal{C}(G, \trianglerighteq), \varphi})$. Thus, (2) fails for some $F, G \in \mathcal{K}(\mathbb{P}, X)$. ■

Fact 1 shows that (2) is too demanding a property. If preferences contain strict comparisons beyond the benchmark order $\succsim$, then stochastic improvements in feasible opportunities need not translate into stochastic improvements in chosen alternatives.

4.2 Comparative Choice Across Preferences

To provide a variation on this theme, suppose now that $\succsim$ corresponds to the preferences of an evaluator who compares the quality of the choices of two people, persons

1 and 2, across all choice problems. Person 1 makes their choices by maximizing a (possibly incomplete) preference relation $\trianglerighteq_1$, and person 2 does the same but using a (possibly incomplete) preference relation $\trianglerighteq_2$. Before the uncertainty is resolved, therefore, the choice behavior of these people on a random menu F is captured by the random sets $\mathcal{C}(F, \trianglerighteq_1)$ and $\mathcal{C}(F, \trianglerighteq_2)$, and the evaluator unambiguously "likes" the choices of person 1 better than those of person 2 if

$$\mathcal{C}(F, \trianglerighteq_1) \succsim^{\text{SD}} \mathcal{C}(F, \trianglerighteq_2). \tag{3}$$

The question is when we would have this for every $F \in \mathcal{K}(\mathbb{P}, X)$. Intuitively, for this to be true, one would think $\trianglerighteq_1$ should be more aligned with $\succsim$ than $\trianglerighteq_2$ is. One way of capturing this idea is: For every $x, y \in X$,

$$x \triangleright_1 y \text{ and } (\text{not } x \succsim y) \quad \text{imply} \quad x \triangleright_2 y. \tag{4}$$

This means when person 1 and the evaluator disagree on the ranking of any two alternatives x and y, person 2 always sides with person 1. In other words, $\trianglerighteq_1$ never rejects a comparison endorsed by $\succsim$ unless $\trianglerighteq_2$ rejects it as well. When this happens, we say that $\trianglerighteq_1$ is **closer to $\succsim$ than** $\trianglerighteq_2$ is.[13]

Fact 2. (3) holds for all $F \in \mathcal{K}(\mathbb{P}, X)$ if, and only if, $\trianglerighteq_1$ is closer to $\succsim$ than $\trianglerighteq_2$ is.

Proof. Suppose $\trianglerighteq_1$ is closer to $\succsim$ than $\trianglerighteq_2$ is, and take any $F \in \mathcal{K}(\mathbb{P}, X)$, and any Borel up-set S in X (relative to $\succsim$). In view of Theorem 2, we will be done if we can show that $\{\mathcal{C}(F, \trianglerighteq_1) \cap S \neq \varnothing\} \supseteq \{\mathcal{C}(F, \trianglerighteq_2) \cap S \neq \varnothing\}$. To this end, take any $\omega \in \Omega$ such that $\mathcal{C}(F(\omega), \trianglerighteq_2) \cap S$ contains at least one element, say, y. If $y \in \mathcal{C}(F(\omega), \trianglerighteq_1)$ we are done, so suppose instead $x \triangleright_1 y$ for some $x \in F(\omega)$. If $x \succsim y$ fails, then $x \triangleright_2 y$ by hypothesis, contradicting $\trianglerighteq_2$-maximality of y in $F(\omega)$. Thus, $x \succsim y$, and this implies $x \in S$ (because S is an up-set). Then, $\mathcal{C}(F(\omega), \trianglerighteq_1) \cap S$ is nonempty because it contains x.

Conversely, suppose there exist $x, y \in X$ such that $x \triangleright_1 y$, (not $x \succsim y$) and (not $x \triangleright_2 y$). Define $F \in \mathcal{K}(\mathbb{P}, X)$ by $F(\omega) := \{x, y\}$, and note that $\mathcal{C}(F, \trianglerighteq_1) = \{x\}$ while $y \in \mathcal{C}(F, \trianglerighteq_2)$. Then, $\mathbb{P}\{\mathcal{C}(F, \trianglerighteq_1) \cap y^{\uparrow} \neq \varnothing\} = 0 < 1 = \mathbb{P}\{\mathcal{C}(F, \trianglerighteq_2) \cap y^{\uparrow} \neq \varnothing\}$. By Theorem 2, (3) fails. ■

Thus, stochastic dominance of the choice behavior of two people reflects how closely one's preferences align with those of the evaluating party. Using $\succsim^{\text{SD}}$ allows us make this sort of a conclusion before the uncertainty underlying the random menus is resolved.

[13] This terminology is not arbitrary. If X is finite, and (4) holds for all $x, y \in X$, the distance between $\trianglerighteq_1$ and $\succsim$ is indeed less than that between $\trianglerighteq_2$ and $\succsim$ according to the Kemeny-Snell-Bogart metric.

4.3 Comparative Statics in Parametric Optimization

The negative nature of Fact 1 stems from imposing (2) for every type of random menus. By contrast, we can use $\succsim^{\text{SD}}$ to make meaningful monotone comparative statics in the standard context of parametric optimization.

First, we recall a few notions from lattice programming. Let $(X, \succsim)$ be a lattice, and recall that any real-valued function g on X is **supermodular** if $g(x \vee y) + g(x \wedge y) \geq g(x) + g(y)$ for all $x, y \in X$. In turn, for any preordered set $(\Theta, \succsim_\Theta)$, we say that a correspondence $\Gamma : \Theta \rightrightarrows X$ is **strongly increasing** if

$$\theta \succsim_\Theta \theta' \quad \text{implies} \quad a \vee b \in \Gamma(\theta) \text{ and } a \wedge b \in \Gamma(\theta') \text{ for every } (a, b) \in \Gamma(\theta) \times \Gamma(\theta').$$

(Obviously, if Γ is strongly increasing, it is increasing.) Finally, a map $f : X \times \Theta \to \mathbb{R}$ is said to have **increasing differences** if

$$x \succsim z \text{ and } y \succsim_\Theta w \quad \text{imply} \quad f(x, y) - f(z, y) \geq f(x, w) - f(z, w) \tag{5}$$

for every $x, z \in X$ and $y, w \in Y$. These concepts are studied thoroughly in Topkis (1998).

FACT 3. *Let $(\Theta, \succsim_\Theta)$ be a partially ordered separable Banach space, and $(X, \succsim)$ a separable Banach lattice. Let $F : \Theta \rightrightarrows X$ be a $\mathcal{B}(\Theta)$-measurable, strongly increasing, compact-valued correspondence. Let $f : X \times \Theta \to \mathbb{R}$ be a function with increasing differences such that $f(\cdot, \theta)$ is continuous and supermodular (for each $\theta \in \Theta$), and $f(x, \cdot)$ is Borel measurable (for each $x \in X$). Then, for any Θ-valued random variables $\boldsymbol{\theta}$ and $\boldsymbol{\theta}'$ on $(\Omega, \Sigma, \mathbb{P})$,*

$$\boldsymbol{\theta} \succsim_{\text{FSD}} \boldsymbol{\theta}' \quad \textit{implies} \quad \arg\max_{x \in F(\boldsymbol{\theta})} f(x, \boldsymbol{\theta}) \succsim^{\text{SD}} \arg\max_{x \in F(\boldsymbol{\theta}')} f(x, \boldsymbol{\theta}').$$

PROOF. Define $M : \Theta \rightrightarrows X$ by $M(\theta) := \arg\max_{x \in F(\theta)} f(x, \theta)$, which is a compact-valued correspondence (by Weierstrass' theorem). By the measurable maximum theorem, $M_{-1}(S) \in \mathcal{B}(\Theta)$ for every closed $S \subseteq X$. (Thus, $M \circ \boldsymbol{\theta} \in \mathcal{K}(\mathbb{P}, X)$ for any Θ -valued random variable on $(\Omega, \Sigma, \mathbb{P})$.) Furthermore, by Topkis' maximum theorem (see Topkis (1978)), M is strongly increasing, whence increasing. Applying Proposition 3 yields the theorem. ■

Fact 3 may be viewed as a stochastic version of Topkis's maximum theorem. Replacing that theorem in the argument with more general monotone maximum theorems (such as that of Milgrom and Shannon (1994)), one may obtain various generalizations of this result.

One can use Fact 3 to obtain stochastic versions of many of the well-known monotone comparative statics results from the literature on strategic complementarities. We conclude this section with a concrete demonstration.

EXAMPLE 10. (*The Law of Random Supply*) Consider a firm that produces a single output by using n inputs. Let $f : \mathbb{R}^n_+ \to \mathbb{R}$ be the production function of the firm. We assume f is increasing and supermodular (which means the inputs are complementary in production). We consider the (parameter) poset $(\Theta, \succsim_\Theta)$ where $\Theta := \mathbb{R}_+ \times \mathbb{R}^n_+$ and $(p, w) \succsim_\Theta (p', w')$ iff $p \geq p'$ and $w' \geq w$. (As usual, the ordering $\geq$ on $\mathbb{R}^n_+$ is the componentwise ordering.) Let us assume that the output and input prices are subject to random shocks; they are given by a Θ-valued random vector $(\mathbf{p},\mathbf{w})$ on $(\Omega, \Sigma,\mathbb{P})$. The (random) **input demand correspondence** of the firm is thus given by the random set $D_{\mathbf{p},\mathbf{w}} : \Omega \rightrightarrows \mathbb{R}^n_+$ with

$$D_{\mathbf{p},\mathbf{w}}(\omega) := \underset{x \in \mathbb{R}^n_+}{\arg\max} \left(\mathbf{p}(\omega) f(x) - \sum_{i=1}^{n} \mathbf{w}_i(\omega) x_i \right).$$

One readily verifies that all the hypotheses of Fact 3 are satisfied in this setup. We may thus conclude that

$$\mathbf{p} \succsim_{\text{FSD}} \mathbf{p}' \text{ and } \mathbf{w}' \succsim_{\text{FSD}} \mathbf{w} \quad \text{imply} \quad D_{\mathbf{p},\mathbf{w}} \succsim^{\text{SD}} D_{\mathbf{p}',\mathbf{w}'},$$

which is but a random version of the classical law of input demand.

Suppose now we fix an input price vector w with $w_i > 0$ for each i. The cost function of the firm is then given by $c : \mathbb{R}_+ \to \mathbb{R}$ with $c(a) := \min\{\sum^n w_i x_i : x \in \mathbb{R}^n_+$ and $f(x) \geq a\}$. In turn, the (random) **supply correspondence** of the firm is the random set $S : \Omega \rightrightarrows \mathbb{R}_+$ with $S_{\mathbf{p}}(\omega) := \arg\max\{\mathbf{p}(\omega)a - c(a) : a \geq 0\}$. We can again apply Fact 3 to conclude that $S_{\mathbf{p}} \succsim^{\text{SD}} S_{\mathbf{p}'}$ whenever $\mathbf{p} \succsim_{\text{FSD}} \mathbf{p}'$. This can aptly be called the *law of random supply* .

5 Properties of $\succsim^{\text{SD}}$

Throughout this section, $(\Omega, \Sigma, \mathbb{P})$ stands for a complete probability space, and $X = (X, \succsim)$ and $Y = (Y, \trianglerighteq)$ properly ordered Polish spaces. When we impose further structure on X, we will be explicit about it.

5.1 Strong Stochastic Dominance Order

The (upper) stochastic dominance order $\succsim^{\text{SD}}$ on $\mathcal{K}(\mathbb{P}, X)$ is not the only reasonable stochastic dominance relation on $\mathcal{K}(\mathbb{P}, X)$ that reduces to the usual stochastic dominance ordering for random variables. In particular, we define the **strong** (upper) **stochastic dominance order** on $\mathcal{K}(\mathbb{P}, X)$ as the binary relation $\succsim^{\text{s-SD}}$ on this set with

$$F \succsim^{\text{s-SD}} G \quad \text{iff} \quad \text{for every } g \in \text{Sel}(G) \text{ there is an } f \in \text{Sel}(F) \text{ with } f \succsim_{\text{FSD}} g.$$

Clearly, $\succsim^{\text{s-SD}}$ is also a preorder on $\mathcal{K}(\mathbb{P}, X)$ that is consistent with the usual (first-order) stochastic dominance order for X-valued random variables. We next show that $\succsim^{\text{SD}}$ is more complete than $\succsim^{\text{s-SD}}$.

PROPERTY I. $\succsim^{\text{s-SD}} \subseteq \succsim^{\text{SD}}$.

PROOF. Take any $F, G \in \mathcal{K}(\mathbb{P}, X)$ with $F \succsim^{\text{s-SD}} G$, and any $\varphi \in C_{b,\uparrow}(X)$. By the measurable maximum theorem, the correspondence $\sigma_G : \Omega \rightrightarrows X$ with $\sigma_G(\omega) := \arg\max\{\varphi(z) : z \in G(\omega)\}$ is Effros measurable and compact-valued. So, by the Kuratowski-Ryll Nardzewski selection theorem, $\text{Sel}(\sigma_G) \neq \varnothing$. Pick any $g \in \text{Sel}(\sigma_G)$, and note that $g \in \text{Sel}(G)$ and $\varphi \circ g =_{\text{a.s.}} s_{G,\varphi}$. Since $F \succsim^{\text{s-SD}} G$, therefore, there is an $f \in \text{Sel}(F)$ with $f \succsim_{\text{FSD}} g$. But then $\mathbb{E}(s_{F,\varphi}) \geq \mathbb{E}(\varphi \circ f) \geq \mathbb{E}(\varphi \circ g) = \mathbb{E}(s_{G,\varphi})$. Since φ was arbitrarily chosen in $C_{b,\uparrow}(X)$, we conclude that $F \succsim^{\text{SD}} G$. ■

It may well be the case that $F \succsim^{\text{SD}} G$, but not $F \succsim^{\text{s-SD}} G$, even when G is single-valued; see Example 1. However, we have $F \succsim^{\text{SD}} G$ iff $F \succsim^{\text{s-SD}} G$, provided that F is single-valued. The following result records a generalization of this observation, and points to the fact that our two stochastic dominance orders on $\mathcal{K}(\mathbb{P}, X)$ coincide in many cases of interest. We relegate its proof to the Appendix.

PROPOSITION 5. *Let $X = (X, \succsim)$ be a separable Banach lattice, and take any $F, G \in \mathcal{K}(\mathbb{P}, X)$ such that $F(\Omega)$ and $G(\Omega)$ are relatively compact, and F is sub-$\vee$-semilattice-valued. Then, $F \succsim^{\text{s-SD}} G$ iff $F \succsim^{\text{SD}} G$.*

5.2 Preservation of $\succsim^{\text{SD}}$ Under Monotone Transformations

We next show that order-preserving transformations preserve the stochastic ordering of random sets.

PROPERTY II. Let $\Phi : X \to Y$ be an order-preserving continuous map. Then for any $F, G \in \mathcal{K}(\mathbb{P}, X)$,

$$F \succsim^{\text{SD}} G \quad \text{implies} \quad \Phi \circ F \succsim^{\text{SD}} \Phi \circ G.$$

PROOF. Take any $F, G \in \mathcal{K}(\mathbb{P}, X)$ with $F \succsim^{\text{SD}} G$, and any $\varphi \in C_{b,\uparrow}(Y)$. It is an easy exercise to verify that $\Phi \circ F, \Phi \circ G \in \mathcal{K}(\mathbb{P}, Y)$, and $s_{\Phi \circ F,\varphi} = s_{F,\varphi \circ \Phi}$ and $s_{\Phi \circ G,\varphi} = s_{G,\varphi \circ \Phi}$. Since $\varphi \circ \Phi \in C_{b,\uparrow}(X)$, therefore,

$$\mathbb{E}(s_{\Phi \circ F,\varphi}) = \mathbb{E}(s_{F,\varphi \circ \Phi}) \geq \mathbb{E}(s_{G,\varphi \circ \Phi}) = \mathbb{E}(s_{\Phi \circ G,\varphi}).$$

In view of the arbitrary choice of φ, we conclude that $\Phi \circ F \succsim^{\text{SD}} \Phi \circ G$. ■

5.3 Preservation of $\succsim^{\text{SD}}$ Under Products and Unions

For any $F \in \mathcal{K}(\mathbb{P}, X)$, recall that $\hat{F}$ is the $k(X)$-valued random variable on $(\Omega, \Sigma, \mathbb{P})$, defined by $\hat{F}(\omega) := F(\omega)$. (See the Remark at the end of Section 3.1.) The push-forward measure induced by $\hat{F}$ on $k(X)$ is denoted by $\mathbb{P}_F$. By definition, $\mathbb{P}_F(\Lambda) := \mathbb{P}\{\omega \in \Omega : F(\omega) \in \Lambda\}$ for any Borel subset Λ of $k(X)$.

For any $F \in \mathcal{K}(\mathbb{P}, X)$ and $F' \in \mathcal{K}(\mathbb{P}, Y)$, we say that F and F' are **independent**, and write $\coprod[F, F']$, if $\hat{F}$ and $\hat{F}'$ are independent random variables on $(\Omega, \Sigma, \mathbb{P})$. This is, of course, the same thing as saying that the probability law of $(\hat{F}, \hat{F}')$ on $k(X) \times k(Y)$ is the product measure $\mathbb{P}_F \otimes \mathbb{P}_{F'}$. Finally, we define the **product** of F and F' as the correspondence $F \times F' : \Omega \rightrightarrows X \times Y$ with $(F \times F')(\omega) := F(\omega) \times F'(\omega)$. This is compact-valued and Effros measurable. (See Molchanov (2017), Theorem 1.3.25.) In what follows, we endow $X \times Y$ with the product topology and order. This makes $X \times Y$ a properly ordered Polish space.

PROPERTY III. For any $F, G \in \mathcal{K}(\mathbb{P}, X)$ and $F', G' \in \mathcal{K}(\mathbb{P}, Y)$ with $\coprod[F, F']$ and $\coprod[G, G']$,

$$F \gtrsim^{\text{SD}} G \text{ and } F' \gtrsim^{\text{SD}} G' \quad \text{imply} \quad F \times F' \gtrsim^{\text{SD}} G \times G'.$$

This result generalizes the corresponding, well-known result for real-valued random variables, and it extends to the case of any finitely many random sets by induction. We relegate its proof to the Appendix.

PROPERTY IV. For any $F, G \in \mathcal{K}(\mathbb{P}, X)$ and $F', G' \in \mathcal{K}(\mathbb{P}, X)$ with $\coprod[F, F']$ and $\coprod[G, G']$,

$$F \gtrsim^{\text{SD}} G \text{ and } F' \gtrsim^{\text{SD}} G' \quad \text{imply} \quad F \cup F' \gtrsim^{\text{SD}} G \cup G'.$$

PROOF. Take any such F, F', G and G', and any Borel up-set S in X. Note that

$$v_{F \cup F'}(S) = v_F(S) + v_{F'}(S) - v_F(S) v_{F'}(S) = 1 - (1 - v_F(S))(1 - v_{F'}(S)),$$

and similarly for $v_{G \cup G'}(S)$. By Theorem 2, $v_F(S) \geq v_G(S)$ and $v_{F'}(S) \geq v_{G'}(S)$, so these formulas yield $v_{F \cup F'}(S) \geq v_{G \cup G'}(S)$. In view of the arbitrary choice of S, and Theorem 2, we conclude $F \cup F' \gtrsim^{\text{SD}} G \cup G'$. ■

Properties III and IV extend inductively to the case of any finitely many random sets by grouping.

5.4 Preservation of $\gtrsim^{\text{SD}}$ Under Random Minkowski Sums

Suppose $X = (X, \gtrsim)$ is a partially ordered separable Banach space. The next property shows that the Minkowski sum of randomly many X-valued random sets stochastically dominates that of another such sum, provided that we have term-by-term dominance, and some natural independence conditions are satisfied.

PROPERTY V. For any Σ-measurable $M : \Omega \to \mathbb{N}$ and $F_1, F_2 ..., G_1, G_2, ... \in \mathcal{K}(\mathbb{P}, X)$ with $\coprod[M, F_1, F_2, ...]$ and $\coprod[M, G_1, G_2, ...]$,

$$F_1 \gtrsim^{\text{SD}} G_1, \ F_2 \gtrsim^{\text{SD}} G_2, \ ... \quad \text{imply} \quad \sum_{i=1}^{M} F_i \gtrsim^{\text{SD}} \sum_{i=1}^{M} G_i.$$

PROOF. Define $F^m := F_1 + \cdots + F_m$ and $G^m := G_1 + \cdots + G_m$ for each $m \in \mathbb{N}$, and put $F := \sum^M F_i$ and $G := \sum^M G_i$. As F and G equal the sum of finitely many nonempty compact sets at each $\omega \in \Omega$, they are compact-valued correspondences. Moreover, by Theorem 1.3.15 of Molchanov (2017), F^m, G^m, F and G are Σ-measurable (for each m). Now take any closed up-set S in X. By Properties II and III, and by induction, we have $v_{F^m}(S) \geq v_{G^m}(S)$ for each $m \in \mathbb{N}$. In addition, by grouping, M and F^m are independent for each m, so

$$v_F(S) = \sum_{m \geq 1} \mathbb{P}\{M = m \text{ and } F^m \cap S \neq \varnothing\} = \sum_{m \geq 1} \mathbb{P}\{M = m\} v_{F^m}(S),$$

and similarly for $v_G(S)$. It follows that $v_F(S) \geq v_G(S)$. In view of Theorem 2 and the arbitrary choice of S, we are done. ■

5.5 Preservation of $\succsim^{\mathbf{SD}}$ Under Weak Convergence

It is well-known that first-order stochastic dominance is preserved under weak convergence. To extend this fact to the context of random sets, take any $F, F_1, F_2, ... \in \mathcal{K}(\mathbb{P}, X)$. In general, we say that (F_m) **weakly converges** to F, and write $F_m \xrightarrow{\text{w}} F$, if $(\hat{F}_m)$ weakly converges to $\hat{F}$ in the usual sense of the term.[14] When X is locally compact, this is equivalent to saying $v_{F_n}(K) \to v_F(K)$ for every $K \in k(X)$ with $v_F(K) = v_F(\text{int}(K))$. And when X is compact, we get the desired conclusion:

PROPERTY VI. Suppose X is compact.[15] For any $F, F_1, ..., G, G_1, ... \in \mathcal{K}(\mathbb{P}, X)$ with $F_m \xrightarrow{\text{w}} F$ and $G_m \xrightarrow{\text{w}} G$,

$$F_1 \succsim^{\text{SD}} G_1,\ F_2 \succsim^{\text{SD}} G_2,\ ... \quad \text{imply} \quad F \succsim^{\text{SD}} G.$$

PROOF. For any such $F, F_1, ..., G, G_1, ...$, and any $\varphi \in C_\uparrow(X)$,

$$\text{C}\int_X \varphi \, \text{d}v_F = \lim \text{C}\int_X \varphi \, \text{d}v_{F_m} \geq \lim \text{C}\int_X \varphi \, \text{d}v_{G_m} = \text{C}\int_X \varphi \, \text{d}v_G$$

where, given that X is compact, the equalities follow from Proposition 1.7.15 of Molchanov (2017). ■

6 Conclusion

We have introduced a notion of first-order stochastic dominance for random sets in ordered Polish spaces, and studied its main properties and applications. Our

[14] We again think of $k(X)$ as endowed with the Hausdorff metric topology here. (Standard texts adopt instead the Fell topology for this purpose, but so long as X is locally compact, these two topologies on $k(X)$ are the same.)

[15] A natural question is if this property holds when X is locally compact as well. We do not know the answer to this at present.

ordering is particularly appealing when the random sets at hand are random menus in an economic environment. It extends the classical stochastic dominance order for random variables, while incorporating the non-additive probabilistic structure induced by random sets through the Choquet integral. We have provided several characterizations of this order, including one in terms of probabilities of hitting upsets, and examined its inheritance under various operations and transformations.

The examples and applications considered in the paper demonstrate that stochastic dominance of random menus exhibits structural features that differ from those of ordinary stochastic dominance. In particular, this order is sensitive to the geometry of feasible sets, and does not admit a Strassen-type coupling characterization. At the same time it remains tractable enough to support meaningful monotone comparative statics analyses in economic contexts, including random budget sets, random choice correspondences, and ex ante parametric optimization.

APPENDIX: Proofs

We begin with introducing a few preliminary notions from the theory of capacities. Let X be a Hausdorff topological space, and v a Borel capacity on X. The **conjugate** of v is the map $v^* : \mathcal{B}(X) \to [0,1]$ with $v^*(S) := 1 - v(X \backslash S)$. This is also a Borel capacity on X, and $v^{**} = v$. Moreover, v is submodular iff v^* is supermodular, while v is continuous at S iff v^* is continuous at $X \backslash S$.

A **Borel probability content** on X is a finitely additive $[0,1]$-valued map μ on $\mathcal{B}(X)$ with $\mu(\varnothing) = 0$ and $\mu(X) = 1$. We denote the collection of all such contents on X by $\Delta_0(X)$. The **core** of v is defined as the set of all Borel probability contents on X that majorize v, that is, $\text{core}(v) := \{\mu \in \Delta_0(X) : \mu \geq v\}$. We can also express the core of v in terms of the conjugate of v as:

$$\text{core}(v) = \{\mu \in \Delta_0(X) : \mu \leq v^*\}. \tag{6}$$

We use this concept to prove the following:

Lemma A. *Let X be a Hausdorff topological space, and v a submodular Borel capacity on X that is continuous at $\varnothing$. Then,*

$$v(S) = \max_{\substack{\mu \in \Delta(X) \\ \mu \leq v}} \mu(S) \quad \text{for every Borel } S \subseteq X.$$

Proof. Take any submodular Borel capacity v on X that is continuous at $\varnothing$. Since v^* is supermodular, Theorem 4.7 of Marinacci and Montrucchio (2004) says $v^*(S) = \min\{\mu(S) : \mu \in \text{core}(v^*)\}$ for every $S \in \mathcal{B}(X)$. On the other hand, as $v^{**} = v$, applying (6) to v^* yields $\text{core}(v^*) = \{\mu \in \Delta_0(X) : \mu \leq v\}$. Consequently,

$$v(S) = 1 - v^*(X \backslash S) = 1 - \min_{\substack{\mu \in \Delta_0(X) \\ \mu \leq v}} \mu(X \backslash S) = \max_{\substack{\mu \in \Delta_0(X) \\ \mu \leq v}} \mu(S)$$

for every $S \in \mathcal{B}(X)$. Next, note that for any $\mu \in \Delta_0(X)$ with $\mu \leq v$, and any sequence (S_m) of Borel subsets of X with $S_m \downarrow \varnothing$, we have $\lim \mu(S_m) \leq \lim v(S_m) = 0$. It follows that every such μ is continuous at $\varnothing$. Since every content on a σ-algebra that is continuous at $\varnothing$ is σ-additive, we see that every $\mu \in \Delta_0(X)$ with $\mu \leq v$ is in fact a Borel probability measure, and the lemma follows. ■

The following is an extension of *Ulam's theorem* on the tightness (= inner regularity) of Borel probability measures to the context of Borel capacities.

PROPOSITION B. *Let X be a Polish space, and v a submodular Borel capacity on X that is continuous at $\varnothing$. Then,*

$$v(S) = \sup_{\substack{K \in k(X) \\ K \subseteq S}} v(K)$$

for every Borel subset S of X.

Proof. Take any $S \in \mathcal{B}(X)$ and $\varepsilon > 0$. By Lemma A, there exists a $\mu \in \Delta(X)$ such that $\mu \leq v$ and $\mu(S) = v(S)$. Since every Borel probability measure on a Polish space is tight (Ulam's theorem), there exists a compact subset K of X with $K \subseteq S$ and $\mu(S) - \mu(K) < \varepsilon$. But then $v(S) - \varepsilon = \mu(S) - \varepsilon \leq \mu(K) \leq v(K)$, and the claim follows. ■

LEMMA C. (The Strict Separation Lemma) *Let $X = (X, \succsim)$ be a properly ordered Polish space. Then, for every closed up-set S in X, there exists a continuous and $\succsim$-increasing map $f : X \to [0,1]$ with $f^{-1}(1) = S$.*

Proof. If $S = X$ the claim is trivial, so we assume $T := X \backslash S$ is nonempty. By hypothesis, $\succsim$ is closed in $X \times X$, and this implies that $x^{\downarrow}$ is closed for any $x \in X$. As $(X, \succsim)$ is normally ordered, the Nachbin separation theorem says that for every $x \in T$ there is a continuous and $\succsim$ -increasing map $f_x : X \to [0,1]$ such that $x^{\downarrow} \subseteq f_x^{-1}(0)$ and $S \subseteq f_x^{-1}(1)$. Define $O_x := \{y \in X : f_x(y) > 0\}$ for each $x \in T$, and note that $\mathcal{O} := \{O_x : x \in T\}$ is an open cover of T. Since X is separable and metrizable, so is T, and it follows that T is Lindelöf. Therefore, there exists a countable subcover of $\mathcal{O}$, that is, there is a sequence (x_m) in T such that $T \subseteq \bigcup_{m \geq 1} O_{x_m}$. Put $f_m := f_{x_m}$ for each $m \geq 1$, and define $f : X \to [0,1]$ by

$$f(y) := \sum_{m=1}^{\infty} 2^{-m} f_m(y).$$

By the Weierstrass M-test, f is continuous. It is also plain that f is $\succsim$ -increasing and $f^{-1}(1) = S$. ■

REMARK. Applying this result to $(X, \precsim)$, we see that for every closed and down-set S in X, there exists a continuous and $\precsim$-decreasing map $f : X \to [0,1]$ with

$f^{-1}(1) = S$. Then, $g := 1 - f$ is a continuous and $\succsim$-increasing $[0,1]$-valued map on X with $g^{-1}(0) = S$.

PROOF OF THEOREM 2. Take any random sets $F, G \in \mathcal{K}(\mathbb{P}, X)$, and assume $F \succsim^{\mathrm{SD}} G$. To prove that (ii) holds, fix a closed up-set S in X. By Lemma C, there exists a continuous and $\succsim$ -increasing map $f : X \to [0,1]$ with $f^{-1}(1) = S$. For each integer $m \geq 1$, define $\varphi_m : X \to [0,1]$ with $\varphi_m(x) := 1 - \min\{m(1 - f(x)), 1\}$. Clearly, $\varphi_m \in C_{b,\uparrow}(X)$ and $\varphi_m|_S = 1$ for each m. Moreover, $\varphi_m \downarrow 1_S$ as $m \to \infty$. Consequently,

$$\begin{aligned} \mathbb{P}\{F \cap S \neq \varnothing\} + \int_{\{F \cap S = \varnothing\}} \max(\varphi_m \circ F)\, \mathrm{d}\mathbb{P} &= \int_\Omega s_{F,\varphi_m}\, \mathrm{d}\mathbb{P} \\ &\geq \int_\Omega s_{G,\varphi_m}\, \mathrm{d}\mathbb{P} \\ &\geq \mathbb{P}\{G \cap S \neq \varnothing\}. \end{aligned}$$

Since $\max \varphi_m(F(\omega)) \downarrow 0$ for every $\omega \in \Omega$ with $F(\omega) \cap S = \varnothing$, applying the monotone convergence theorem yields $\mathbb{P}\{F \cap S \neq \varnothing\} \geq \mathbb{P}\{G \cap S \neq \varnothing\}$, as desired.

Now assume (ii). Take any Borel up-set S in X, and fix an arbitrary $\varepsilon > 0$. By Proposition B, there exists a compact subset K of X such that $K \subseteq S$ and $v_G(S) - v_G(K) < \varepsilon$. We claim that $K^\uparrow$ is closed. To see this, take any sequence (x_m) in $K^\uparrow$ with $x_m \to x$ for some $x \in X$. Then, there is a sequence (y_m) in K such that $x_m \succsim y_m$ for each m. Since K is compact, there is a subsequence (y_{m_k}) of (y_m) which converges in K. As $\succsim$ is closed in $Y \times Y$, we have $x \geq \lim y_{m_k}$, and it follows that $x \in K^\uparrow$. Conclusion: $K^\uparrow$ is closed. Since $K \subseteq K^\uparrow \subseteq S$, we may thus apply (ii) to find

$$v_F(S) \geq v_F(K^\uparrow) \geq v_G(K^\uparrow) \geq v_G(K) > v_G(S) - \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary here, therefore, $v_F(S) \geq v_G(S)$. We proved: (ii) implies (iii).

Since (iv) implies (i) trivially, it remains to prove that (iii) implies (iv). To this end, assume $\mathbb{P}\{F \cap S \neq \varnothing\} \geq \mathbb{P}\{G \cap S \neq \varnothing\}$ for every Borel up-set S in X. Take an arbitrary $\varphi \in L_{b,\uparrow}(X)$, and note that $\{\varphi \geq a\}$ is a Borel up-set in X for any $a > 0$. Put $\alpha := \inf \varphi(X)$, and fix an arbitrary $m \in \mathbb{N}$. Since φ is bounded, there exists a real number k_m with $\alpha + k_m m^{-1} > \sup \varphi(X)$. We define $S_i := \{\varphi \geq \alpha + im^{-1}\}$ for each $i = 1, ..., k_m$, and consider the map

$$\varphi_m := \alpha + m^{-1} \sum_{i=0}^{k_m} 1_{S_i}.$$

For each $i = 1, ..., k_m$, consider the Σ-measurable sets

$$A_i := \{\omega \in \Omega : \varphi_m(z) \geq \alpha + im^{-1} \text{ for some } z \in F(\omega)\}.$$

Then, $\Omega = A_1 \supseteq \cdots \supseteq A_{k_m}$, and

$$s_{F,\varphi_m}(\omega) = \alpha + im^{-1} \quad \text{for all } \omega \in A_i \backslash A_{i+1}$$

for all $i = 1, ..., k_m - 1$, while $s_{F,\varphi_m}(\omega) = \alpha + k_m m^{-1}$ for all $\omega \in A_{k_m}$. Consequently,

$$\begin{aligned}\int_\Omega s_{F,\varphi_m} \mathrm{d}\mathbb{P} &= \sum_{i=1}^{k_m-1} \left(\alpha + im^{-1}\right) \mathbb{P}(A_i \backslash A_{i+1}) + \left(\alpha + k_m m^{-1}\right) \mathbb{P}(A_{k_m}) \\ &= \left(\alpha + m^{-1}\right) + m^{-1} \sum_{i=2}^{k_m} \mathbb{P}(A_i).\end{aligned}$$

Now for each $i = 1, ..., k_m$, define

$$B_i := \{\omega \in \Omega : \varphi_m(z) \geq \alpha + im^{-1} \text{ for some } z \in G(\omega)\},$$

and note that, analogously,

$$\int_\Omega s_{G,\varphi_m} \, \mathrm{d}\mathbb{P} = \left(\alpha + m^{-1}\right) + m^{-1} \sum_{i=2}^{k_m} \mathbb{P}(B_i).$$

But $A_i = F_{-1}(S_i)$ and $B_i = G_{-1}(S_i)$, so by hypothesis, we have $\mathbb{P}(A_i) \geq \mathbb{P}(B_i)$, for each $i = 1, ..., k_m$. It thus follows from the computations above that $\mathbb{E}(s_{F,\varphi_m}) \geq \mathbb{E}(s_{G,\varphi_m})$. Since $m \in \mathbb{N}$ is arbitrary here, and $\varphi_m \to \varphi$ uniformly, we conclude that $\mathbb{E}(s_{F,\varphi}) \geq \mathbb{E}(s_{G,\varphi})$. The proof of Theorem 2 is now complete. ■

For any complete measurable space (Ω, Σ) and Polish space X, the projection of any $A \in \Sigma \otimes \mathcal{B}(X)$ to Ω, that is,

$$\pi_\Omega(A) := \{\omega \in \Omega : (\omega, x) \in A \text{ for some } x \in X\},$$

is a Σ-measurable set. This is the classical *Measurable Projection Theorem* .[16] Using this theorem we now prove a gap that was left in the main exposition (specifically in Example 9 and Section 4.2).

THEOREM D. (Measurability of the Closed Maximal Correspondence) *Let $(\Omega, \Sigma, \mathbb{P})$ be a complete probability space, and X a Polish space. For any closed preorder $\succsim$ on X, and $F \in \mathcal{K}(\mathbb{P}, X)$, define $M : \Omega \to 2^X$ by*

$$M(\omega) := \mathrm{cl}(\mathrm{MAX}(F(\omega), \succsim)).$$

If either (i) X is compact; or (ii) $\succsim$ is a partial order, $M \in \mathcal{K}(\mathbb{P}, X)$.

Proof. As noted in Example 9, we only need to prove the Effros measurability of M. To simplify the notation, let us define the correspondence $M_0 : \Omega \rightrightarrows X$ by $M_0(\omega) := \mathrm{MAX}(F(\omega), \succsim)$. Now fix an arbitrary open set O in X, and note that $O \cap M(\omega) \neq \varnothing$

[16] This is a special case of Theorem III.23 of Castaing and Valadier (1977), who call it the *projection theorem* . The latter result is in fact more general in that it takes X as a Souslin space (= a continuous image of a Polish space).

iff $O \cap M_0(\omega) \neq \varnothing$ (because an open set that intersects the closure of a set, must intersect the set itself). Our task is then to show that $\{M_0 \cap O \neq \varnothing\} \in \Sigma$. But $\{M_0 \cap O \neq \varnothing\} = \pi_\Omega(\mathrm{Gr}(M_0) \cap (\Omega \times O))$, so in view of the measurable projection theorem, it is enough to show that $\mathrm{Gr}(M_0) \in \Sigma \otimes \mathcal{B}(X)$.

In what follows we consider $k(X)$ as a metric space relative to the Hausdorff metric d^{H}, and recall that Effros measurability of F is actually equivalent to say that the map $\omega \mapsto F(\omega)$ is $\Sigma \backslash \mathcal{B}(X)$-measurable; see Theorem 18.10 of Aliprantis and Border (2006).

Consider now the case (i): X is compact. In this case, by Theorem 3 of Nishimura and Ok (2016), there exists a nonempty compact $\mathcal{U} \subseteq C(X)$ such that $x \succsim y$ iff $u(x) \geq u(y)$ for all $u \in \mathcal{U}$. We define $H : X \times X \to \mathbb{R}$ by

$$H(x, y) := \max_{u \in \mathcal{U}} (u(y) - u(x)).$$

As $\mathcal{U}$ is compact, this is a (well-defined) continuous function. (Indeed, $(u, x, y) \mapsto u(y) - u(x)$ is readily checked to be continuous for all $u \in \mathcal{U}$ and $x, y \in X$, hence given that $\mathcal{U}$ is compact, Berge's maximum theorem applies.) For any $\varepsilon > 0$, let A_ε be the set of all $(K, x) \in k(X) \times X$ such that (i) $x \in K$; and (ii) there exists a $y \in K$ with $H(y, x) \leq 0$ and $H(x, y) \geq \varepsilon$. This set is closed in $k(X) \times X$ for any $\varepsilon > 0$. (To see this, take any $(K, x), (K_1, x_1), (K_2, x_2), ... \in k(X) \times X$ such that $d^{\mathrm{H}}(K_n, K) \to 0$, $x_n \to x$, and $(K_n, x_n) \in A_\varepsilon$ for all $n \geq 1$. Then, $x_n \in K_n$ for each n, so $x \in K$. Moreover, for each n, there is a $y_n \in K_n$ with $H(y_n, x_n) \leq 0$ and $H(x_n, y_n) \geq \varepsilon$. As X is compact, we can assume that $y_n \to y$ for some $y \in X$, passing to a subsequence if necessary. Since $d^{\mathrm{H}}(K_n, K) \to 0$ and $y_n \in K_n$ for each n, we must have $y \in K$. By continuity of H, $H(y, x) \leq 0$ and $H(x, y) \geq \varepsilon$.) As the map $(\omega, x) \mapsto (F(\omega), x)$ is clearly $(\Sigma \otimes \mathcal{B}(X)) \backslash (\mathcal{B}(k(X)) \times X)$-measurable, therefore,

$$B_m := \{(\omega, x) \in \Omega \times X : (F(\omega), x) \in A_{m^{-1}}\} \in \Sigma \otimes \mathcal{B}(X)$$

for each $m \in \mathbb{N}$. We put $B := \bigcup_{m \geq 1} B_m$. Then, $B \in \Sigma \otimes \mathcal{B}(X)$, while $(\omega, x) \in B$ iff $x \notin M_0(\omega)$. It follows that $\mathrm{Gr}(M_0) = \mathrm{Gr}(F) \backslash B \in \Sigma \otimes \mathcal{B}(X)$, as we sought.

We now take on the case (ii). Pick any metric d on X that (completely) metrizes the topology of X, and define $D : \Omega \times X \to \mathbb{R}$ by

$$D(\omega, x) := \begin{cases} \max\{d(x, y) : y \in F(\omega) \cap x^{\uparrow}\}, & \text{if } x \in F(\omega) \\ -1, & \text{otherwise.} \end{cases}$$

This is well-defined because $F(\omega)$ is compact for all $\omega \in \Omega$ and $x^{\uparrow}$ is closed for all $x \in X$. Moreover, for every $\omega \in \Omega$, we have $x \in M_0(\omega)$ iff $D(\omega, x) = 0$ (because $\succsim$ is antisymmetric). Thus, $\mathrm{Gr}(M_0) = D^{-1}(0)$, so it is enough to show that D is $\Sigma \otimes \mathcal{B}(X)$-measurable. To do this, we consider the map $\phi : k(X) \times X \to \mathbb{R}$ with

$$\phi(K, x) := \begin{cases} \max\{d(x, y) : y \in K \cap x^{\uparrow}\}, & \text{if } K \cap x^{\uparrow} \neq \varnothing \\ -1, & \text{otherwise.} \end{cases}$$

Obviously, $D(\omega,x)=\phi(F(\omega),x)$ for all $\omega\in\Omega$ and $x\in X$. As $(\omega,x)\mapsto(F(\omega),x)$ is a $(\Sigma\otimes\mathcal{B}(X))\backslash(\mathcal{B}(k(X))\times X)$-measurable map on $\Omega\times X$, therefore, we will be done if we can show that ϕ is measurable. To this end, take any $t>-1$, and note that $\{\phi>t\}=\bigcup_{m\geq 1}C_m$ where

$$C_m:=\{(K,x)\in k(X)\times X:\ \text{there is a } y\in K\cap x^{\uparrow} \text{ with } d(x,y)\geq t+m^{-1}\}$$

for each $m\in\mathbb{N}$. We will conclude our proof by showing that C_m is closed in $k(X)\times X$ for each m. So fix any $m\in\mathbb{N}$, set $C:=C_m$, and take any $(K,x),(K_1,x_1),(K_2,x_2),...\in k(X)\times X$ such that $d^{\mathrm{H}}(K_n,K)\to 0$, $x_n\to x$, and $(K_n,x_n)\in C$ for all $n\geq 1$. Then, for each n, there is a $y_n\in K_n$ with $y_n\succsim x_n$ and $d(x_n,y_n)\geq t+m^{-1}$. But as $K_n\to K$ relative to the Hausdorff metric, and K is compact, the sequence (y_n) is eventually contained in a common compact set, hence admits a covergent subsequence. Relabeling if necessary, we denote this subsequence also as (y_n), and put $y:=\lim y_n$. As $d(y_n,K)\leq d^{\mathrm{H}}(K_n,K)\to 0$, and $d(\cdot,K)$ is continuous, we have $d(y,K)=0$, that is, $y\in K$. Moreover, $y\succsim x$ (because $\succsim$ is closed), and $d(x,y)\geq t+m^{-1}$ (because d is continuous). Thus $(K,x)\in C$, and we conclude that C is closed. ■

Proof of Proposition 5. Set $Y:=\mathrm{cl}(F(\Omega)\cup G(\Omega))$ which is a compact subspace of X. Then, $C(Y)$, and hence $C_{\uparrow}(Y)$, are separable metric spaces. Take a countable dense set in $C_{\uparrow}(Y)$, and enumerate it as $\{\varphi_1,\varphi_2,...\}$.

Now suppose $F\succsim^{\mathrm{SD}}G$, and fix any $g\in\mathrm{Sel}(G)$. By Proposition 1, for each positive integer i there is an $f_i\in\mathrm{Sel}(F)$ with $\mathbb{E}(\varphi_i\circ f_i)\geq\mathbb{E}(\varphi_i\circ g)$. For each $m\in\mathbb{N}$, we define $h_m:\Omega\to Y$ by $h_m(\omega):=f_1(\omega)\vee\cdots\vee f_m(\omega)$. Since F is sub-$\vee$-semilattice-valued, $h_m(\omega)\in F(\omega)$ for all $m\in\mathbb{N}$ and $\omega\in\Omega$. Moreover, for each $\omega\in\Omega$,

$$\cdots\succsim h_2(\omega)\succsim h_1(\omega) \tag{7}$$

while, since $F(\omega)$ is compact, there is a strictly increasing sequence (m_k) in $\mathbb{N}$ such that $h_{m_k}(\omega)\to x_\omega$ for some $x_\omega\in F(\omega)$. Then, for every $\omega\in\Omega$ and $n\in\mathbb{N}$ there is a $K\in\mathbb{N}$ such that $h_{m_k}(\omega)\succsim h_n(\omega)$ for every $k\geq K$, so letting $k\to\infty$ yields $x_\omega\succsim h_n(\omega)$ (because $\succsim$ is closed in $X\times X$). Now define $f:\Omega\to Y$ by $f(\omega):=x_\omega$. Given that $(X,\succsim)$ is a Banach lattice, (7) implies $f(\omega)=\lim h_m(\omega)$ for all $\omega\in\Omega$, so f is Σ-measurable. Thus: $f\in\mathrm{Sel}(F)$ and $f\succsim f_m$ for all $m\in\mathbb{N}$. The latter property implies $\varphi_i\circ f\geq\varphi_i\circ f_m$ for all $i,m\in\mathbb{N}$. Consequently, $\mathbb{E}(\varphi_i\circ f)\geq\mathbb{E}(\varphi_i\circ f_i)\geq\mathbb{E}(\varphi_i\circ g)$ for all $i\in\mathbb{N}$. As $\{\varphi_1,\varphi_2,...\}$ is dense in $C_{\uparrow}(Y)$, this implies $f\succsim_{\mathrm{FSD}}g$. As g was arbitrarily chosen in $\mathrm{Sel}(G)$, we conclude that $F\succsim^{\text{s-SD}}G$. In view of Property I, our proof is complete. ■

Proof of Property III. Take any F,G,F' and G' with the posited properties. Fix an arbitrary closed up-set C in $X\times Y$. By Theorem 2, we will be done if we can show that

$$\mathbb{P}\{(F\times F')\cap C\neq\varnothing\}\geq\mathbb{P}\{(G\times G')\cap C\neq\varnothing\}.$$

To this end, we define

$$S := \{(A, B) \in k(X) \times k(Y) : (A \times B) \cap C \neq \varnothing\}.$$

For any $A \in k(X)$ and $B \in k(Y)$, we denote the sections of S at A and at B by S_A and S^B, that is, $S_A := \{B \in k(X) : (A, B) \in S\}$ and $S^B := \{A \in k(Y) : (A, B) \in S\}$. We set $C(B) := \{x \in X : (x, y) \in C \text{ for some } y \in B\}$, and note that $\{(F \times B) \cap C \neq \varnothing\} = \{F \cap C(B) \neq \varnothing\}$, for any $B \in k(Y)$. But then

$$\mathbb{P}_F(S^B) = \mathbb{P}\{F \in S^B\} = \mathbb{P}\{(F, B) \in S\} = \mathbb{P}\{(F \times B) \cap C \neq \varnothing\} = \mathbb{P}\{F \cap C(B) \neq \varnothing\},$$

and similarly, $\mathbb{P}_G(S^B) = \mathbb{P}\{G \cap C(B) \neq \varnothing\}$, for any $B \in k(Y)$. But $C(B)$ is a closed up-set in X for any $B \in k(Y)$. Thus, by Theorem 2, $F \gtrsim^{\text{SD}} G$ implies $\mathbb{P}_F(S^B) \geq \mathbb{P}_G(S^B)$ for any $B \in k(Y)$. Integrating both sides of this inequality against $\mathbb{P}_{F'}$ yields

$$\mathbb{P}_F \otimes \mathbb{P}_{F'}(S) = \int_{k(Y)} \mathbb{P}_F(S^B)\, \mathbb{P}_{F'}(\, \mathrm{d}B) \geq \int_{k(Y)} \mathbb{P}_G(S^B)\, \mathbb{P}_{F'}(\, \mathrm{d}B).$$

In turn, reasoning analogously with the section S_A, we find $\mathbb{P}_{F'}(S_A) \geq \mathbb{P}_{G'}(S_A)$ for any $A \in k(X)$, and integrating this inequality against $\mathbb{P}_G$ yields

$$\int_{k(X)} \mathbb{P}_{F'}(S_A)\, \mathbb{P}_G(\, \mathrm{d}A) \geq \int_{k(X)} \mathbb{P}_{G'}(S_A)\, \mathbb{P}_G(\, \mathrm{d}A) = \mathbb{P}_G \otimes \mathbb{P}_{G'}(S)$$

by Tonelli's theorem. Again by Tonelli's theorem, we can combine these inequalites to conclude that

$$\begin{aligned} \mathbb{P}\{(F \times F') \cap C \neq \varnothing\} &= \mathbb{P}_F \otimes \mathbb{P}_{F'}\{(A, B) : (A \times B) \cap C \neq \varnothing\} \\ &\geq \mathbb{P}_G \otimes \mathbb{P}_{G'}\{(A, B) : (A \times B) \cap C \neq \varnothing\} \\ &= \mathbb{P}\{(G \times G') \cap C \neq \varnothing\} \end{aligned}$$

where we used the independence of F and F' to get the first equality, and that of G and G' to get the second. ■

References

Beresteanu, A., I. Molchanov, and F. Molinari, Sharp identification regions in models with convex moment predictions, *Econometrica* 79 (2011), 1785-1821.

Beresteanu, A., I. Molchanov, and F. Molinari, Partial identification using random set theory, *J. Econometrics* 166 (2012), 17-32.

Cerreia-Vioglio, S., and E. A. Ok, Abstract integration of set-valued functions, *J. Math. Anal. Appl.* 547 (2025), Article: 129169.

Castaing, C., and M. Valadier, *Convex Analysis and Measurable Multifunctions*, Springer Berlin, Heidelberg, 1977.

Fernández, I., and I. Molchanov, A stochastic order for random vectors and random sets based on the Aumann expectation, *Stat. Probab. Lett.* 63 (2003), 295-305.

Kamae, T., U. Krengel, and G. O'Brien, Stochastic inequalities on partially ordered spaces, *Ann. Probab.* 5 (1977), 899-912.

Kuratowski, K., and C. Ryll-Nardzewski, A general theorem on selectors, *Bull. Acad. Pol. Sci.* 13 (1965), 397-403.

Marinacci, M., and L. Montrucchio, Introduction to the mathematics of ambiguity, in: I. Gilboa (Ed.), *Uncertainty in Economic Theory*, Routledge, New York, 2004.

Marshall, A., Multivariate stochastic orderings and generating cones of functions, in *Stochastic Orders and Decision under Risk*, IMS Lecture Notes - Monograph Series, 1991.

Milgrom, P., and C. Shannon, Monotone comparative statics, *Econometrica* 62 (1994), 157-180.

Molchanov, I., *Theory of Random Sets*, 2nd ed., Springer, London, 2017.

Molchanov, I., and F. Molinari, *Random Sets in Econometrics*, Cambridge University Press, Cambridge, 2018.

Mosler, K.. and M. Scarsini, Some theory of stochastic dominance, in *Stochastic Orders and Decision under Risk*, IMS Lecture Notes - Monograph Series, 1991.

Nishimura, H., and E. A. Ok, Utility representation of an incomplete and nontransitive preference relation, *J. Econom. Theory* 166 (2016), 164-185.

Topkis, D. , Minimizing a submodular function on a lattice. *Oper. Res.* 26 (1978), 305-321.

Topkis, D., *Supermodularity and Complementarity*, Princeton University Press, Princeton, 1998.